☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner:	County of Hawaii		
Contractor:	Nan, Inc.	**Project No.:**	WW-4705R
Project Name:	Hilo WWTP Phase 1	**Submittal Number:**	11348-001.0
Submittal Title:	Chemical Mixers	☐ For Information Only	
TO:	County		
From:	Nan Inc.		

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 11348	**Paragraph:** 1.04, 1.05, 2.00		
Authored By: Evoqua Water Technologies		**Date Submitted:**	9/29/2025

Submittal Certification

Check Either (A) or (B):

☐ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☑ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: Marshall Rucknagel

Printed Name and Title: Marshall Rucknagel, Project Engineer

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:

Date Received PM/CM to Reviewer:

Date Received Reviewer to PM/CM:

Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/29/2025
SPECIFICATION SECTION # 11348
SPECIFICATION Chemical Mixers
PARAGRAPH 1.04, 1.05, 2.00
DRAWING N/A
SUBCONTRACTOR N/A
SUPPLIER EVOQUA
MANUFACTURER N/A

CERTIFIED BY CQCM or Designee :

New Section

SECTION 11348[AD4]

CHEMICAL MIXERS

PART 1 GENERAL

1.01 SUMMARY

A. Section includes: Chemical induction mixer (CIM) to inject sodium hypochlorite (SHC) to chlorinate and disinfect secondary effluent at the Chlorine Contact Tank for the production of No. 3 Water (3W).

B. As specified in Section 15050 - Common Work Results for Mechanical Equipment.

C. Tag numbers:
   1. 3W Chlorine Mixer 09-MIX-1110.

1.02 REFERENCES

A. ASTM International (ASTM):
   1. A276 - Standard Specification for Stainless Steel Bars and Shapes.

B. CSA International (CSA).

C. National Electrical Manufacturers Association (NEMA):
   1. 250 - Enclosure for Electrical Equipment (1000 V Maximum).

D. Underwriters Laboratories, Inc. (UL).

1.03 DEFINITIONS

A. NEMA:
   1. Type 4X enclosure in accordance with NEMA 250.

1.04 SYSTEM DESCRIPTION

A. CIMs shall be designed to be mounted in a concrete tank downstream of the Effluent Flume and in the Chlorine Contact Tank, as indicated on the Drawings.

B. Capable of injecting SHC into the water stream at a velocity of 60 feet per second without diluting with water overflow range as scheduled.

See exception 1

C. Capable of passing solids up to 0.5-inch in diameter or stringy materials that may be present in the water.

D. Manufacturer shall be responsible for the structural design of each mixer frame and support system. The manufacturer shall evaluate the velocity conditions and pressure at the installation location and design support framing, cables, and flexible hose for these conditions.

See exception 2

E. Furnish a complete system for each CIM, including motor, variable speed drive, chemical piping or hosing, packing gland assembly, Type 316 stainless steel mounting flange, mounting rail with insertion and retraction device, and the necessary mounting brackets and all other necessary appurtenances. Coordinate details with dimensions indicated on the Drawings.

F. All chemical mixer unit components, accessories, and spare parts shall be provided by the same manufacturer.

G. Design requirements: CIM capable of handling the following fluid as scheduled:
 1. SHC:
  a. Dry chemical formula: NaOCl.
  b. Solution concentration: 12.5 percent by weight.
  c. Solution pH: 11 to 13.
  d. Solution specific gravity: 1.1 to 1.2.

1.05 SUBMITTALS

A. Submit as specified in Section 01330 - Submittal Procedures.

B. Product data: As specified in Section 15050 - Common Work Results for Mechanical Equipment.

C. Shop drawings: As specified in Section 15050 - Common Work Results for Mechanical Equipment.

Will be submitted under a separate cover

D. Calculations: As specified in Section 15050 - Common Work Results for Mechanical Equipment:
 1. Demonstrating compliance with structural criteria and requirements of this Section.
 2. Substantiating continuous running torque rating, thermal rating, and overload torque rating of each major drive train component.
 3. Bearing life requirements.

Will be submitted under a separate cover

E. Vendor operation and maintenance manuals: As specified in Section 01782 - Operation and Maintenance Manuals.

Will be submitted under a separate cover

F. Commissioning submittals:
 1. Provide Manufacturer's Certificate of Source Testing as specified in Section 01756 - Commissioning.
 2. Provide Manufacturer's Certificate of Installation and Functionality Compliance as specified in Section 01756 - Commissioning.

1.06 WARRANTY

A. Provide warranty as specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. One of the following or equal:
   1. Evoqua, "Water Champ", Model SWC2F Chemical Induction Unit.

B. Manufacturer shall coordinate the length of the power cable, hoist cables, mounting system, vacuum tubing, and all other pertinent items with the elevations as indicated on the Drawings.

2.02 MATERIALS

A. Piping connection: Grade 2 (unalloyed) titanium or PVC.

B. Propeller: An exposed four-bladed airfoil design made of Grade 2 (unalloyed) titanium or Hastelloy C.

C. Shaft: Grade 2 (unalloyed) titanium.

D. Motor shaft: Titanium or stainless steel with titanium sleeve.

E. Vacuum enhancer: Material suitable for intended service.

F. Vacuum chamber/housing: Grade 2 (unalloyed) titanium.

G. Wetted parts: All wetted parts of the inductor system, or material coming in contact with non-diluted chemical shall be made of Grade 2 (unalloyed) titanium.

See exception 3 H. Orifice housing: Grade 2 (unalloyed) titanium or High-density PVC.

I. Guide rail: ASTM A276, Type 316 stainless steel.

See exception 4 J. Lifting cable: ASTM A276, Type 316 stainless steel.

K. Mounting bolts, nuts, and washers: ASTM A276, Type 316 stainless steel.

See exception 4, 5 L. Flange mounting: ASTM A276, Type 316 stainless steel.

See exception 4 M. Crane, cable and mounting socket: ASTM A276, Type 316 stainless steel.

N. Hand operated winch: ASTM A276, Type 316 stainless steel or galvanized steel.

See exception 5 O. Seal plate: Titanium.

P. Motor housing: ASTM A276, Type 316 stainless steel.

2.03 COMPONENTS

A. Motors:
   1. Motor: Type 316 stainless steel construction with a hermetically sealed, externally water-cooled enclosure.

2. Moisture: Resistant Class B insulation and thermal overload protection.

See exception 6 3. NEMA Design B designed for continuous duty in a submerged environment and capable of running dry for an extended period of time without damage.

4. Suitable for 460-volt, 3-phase, 60-Hz power.

5. Motor Windings: Hermetically sealed and resin-filled in an all-welded Type 316 stainless steel shell for moisture resistance.

6. The motor shall have an integral designed shaft with a mechanical seal using carbon versus ceramic seal faces. Motor shaft shall be supported by two permanent lubricated ball bearings with a minimum L10 life of 50,000 hours.

See exception 7 7. The electrical cable entranceway to each motor shall be provided with positive strain relief to prevent leakage or pull-out of the cable in the event that a force is accidentally placed on the cable during the lowering of the inductor unit.

See exception 8 8. Incoming lead wires shall be spliced in the motor terminal housing. After splicing, the terminal housing shall be filled with epoxy to seal the outer cable jacket and the individual strands to prevent all possibility of water entering the motor housing or the terminal housing. A secondary elastomer compression grommet shall also be supplied. The combination of the epoxy seal and compression grommet shall provide complete seal ring and strain relief.

See exception 8 9. Sufficient cable shall be supplied to extend from the motors to the induction unit controls without splicing. A quick-disconnect type coupling shall be provided for each cable at surface junction box.

10. The motors, cables, and electronic controls shall be sized, furnished, and installed so that the motors shall never exceed the nameplate rating.

B. Power cables:

1. The CIM shall be furnished complete with power cable.
2. Cable shall have UL and/or CSA approved insulation.

2.04 ACCESSORIES

A. Guide rails, supports, and lifting device:

1. Type: Suitable for installation as scheduled below and as indicated on the Drawings.

2. Materials:

See exception 4 a. Wet Pit: Type 316 stainless steel guide rails, lifting cable or chain and wall supports; Type 316 stainless steel cage assembly, where applicable; Type 316 stainless steel anchor bolts and fasteners.

3. Wet Pit Guide Rails:

a. Type: Single or dual pipe or rail able to accurately guide the CIM.
b. Strength: Withstand a minimum of 1.5 times the maximum imposed operating loads or 1.0 times the imposed Uniform Building Code seismic loads, whichever is greater.
c. Intermediate Supports: Provide at 5-foot maximum intervals; less as required to provide support.

See exception 4 4. Lifting Device: Provide portable equipment davit crane as specified in Section 14612 – Davit Cranes and as follows:

a. Cable Length: Able to lower CIM from top of concrete tank or wet well to operating position as indicated on the Drawings plus an additional 5 feet of length.
b. Base Mount: 316 SST wall-mount type.

B. Mounting bolts:
   1. Provide Type 316 stainless steel mounting bolts of the size as recommended by the manufacturer.
   2. Provide stamped and signed anchorage calculations for guide rail, supports and lifting device.

2.05 VENDOR CONTROL PANEL VCP

A. Provide control system with equipment protective devices for safe operation in a local control panel and with remote inputs and outputs as indicated on the Drawings and as specified.

B. Each CIM shall be provided with a pre-wired NEMA 4X VCP:
   1. The VCP shall be sized to match the voltage and horsepower of the motor to be controlled.
   2. Provide HAND/OFF/AUTO selector switch and START, LOCKOUT/STOP pushbuttons for control:
      a. When in HAND, the unit shall start and stop from the start/stop pushbuttons.
      b. When in OFF, the unit shall not run.
      c. When in AUTO, the unit shall be controlled by a discrete RUN signal from the plant SCADA.
   3. Provide the following push-to-test pilot lights:
      a. Red RUNNING.
      b. Amber FAILED.
      c. White POWER ON.
   4. A common fault alarm condition shall require manual depression of the VCP RESET pushbutton before the CIM may be re-started. The VCP shall be provided with internal "seal-in" logic to accomplish this feature.
   5. VCP and all components shall meet the requirements of Section 17710 - Control Systems - Panels, Enclosures, and Panel Components.
   6. Provide motor starter per Section 16422 - Motor Starters.

C. Each control panel shall monitor the CIM and shall shut the unit down if any of the following conditions occur:
   1. Overcurrent.
   2. Undercurrent.
   3. Motor high temperature.
   4. Rapid cycling.

D. Provide 3-phase, 460-volt power with lockable disconnect switch. Disconnect rated for 600 volts.

E. Run time meter up to 99,999 hours.

F. Provide dry contacts for the following:
   1. RUNNING.
   2. FAILED.
   3. IN AUTO.

G. In remote mode, unit shall be capable of remote start/stop from the PLC as indicated on the Drawings.

2.06 SPARE PARTS

A. Provide the following spare parts for each CIM:
   1. 25 feet long, 3/4-inch diameter hose assembly with Camlock fittings.
   2. 4-inch vacuum enhancer for 4-inch motors.
   3. Enhancer set screws (2 per CIM).
   4. Special tools. See exception 9
   5. Spare lab seal for 4-inch vacuum body.
   6. Spare seals for 1-1/2-inch Camlock fittings, Viton (2 per hose).

PART 3 EXECUTION

3.01 INSTALLATION

A. Install unit in accordance with the manufacturer's instructions and as indicated on the Drawings.

3.02 COMMISSIONING

A. As specified in Section 01756 - Commissioning and this Section.

B. Manufacturer services:
   1. Provide certificates:
      a. Manufacturer's Certificate of Source Testing.
      b. Manufacturer's Certificate of Installation and Functionality Compliance.
   2. Manufacturer's Representative onsite requirements:
      a. Installation: 1 trip, 2-day minimum.
      b. Functional Testing: 1 trip, 2-day minimum each.
   3. Training:
      a. Maintenance: 4 hours per session, 2 sessions.
      b. Operation: 2 hours per session, 2 sessions.
   4. Process operational period:
      a. As required by Owner.

C. Source testing:
   1. Test as specified in Section 15958 - Mechanical Equipment Testing.
   2. Electrical Instrumentation and Controls:
      a. Test witnessing: Not witnessed.
      b. Conduct testing as specified in Section 17950 - Commissioning for Instrumentation and Controls.

D. Functional testing:
   1. Mixer:
      a. Test witnessing: Witnessed.
      b. Conduct Level 1 General Equipment Performance Test:
         1) Verify that the CIM can achieve the required feed rates.
   2. Electrical Instrumentation and Controls:
      a. Test witnessing: Witnessed.
      b. Conduct testing as specified in Section 17950 - Commissioning for Instrumentation and Controls.

3.03 SCHEDULE

A. CIMs shall be supplied as specified and in accordance with the following schedule:
   1. General characteristics:

Tag Number	**09-MIX-1110**
Agency	Hilo WWTP
Location	At 16-inch inlet to Chlorine Contact Tank from Effluent Flume
Quantity	1
Service	SHC Solution at 12.5 percent
Installation Type	Standard Rail Assembly
Removal Mechanism	Davit Crane
Feed (gal/day)	50-500
Max Liquid Induction (gpm)	5
Max Vacuum (in. Hg)	22

   2. Motor characteristics:

Tag Number	**09-MIX-1110**
Minimum Motor Horsepower (hp)	2
Service Factor	1.15
Motor Voltage (volt)	460
Power Phases	3
Frequency (Hertz)	60
Motor Enclosure	Submersible

END OF SECTION

[AD4] Addendum No. 4 - June 2024

Use one of the cells to the right for a Spine Tab.

First Column is 1” wide for a 1 ½” Binder

Second Column is 1 ½” wide for a 2” Binder

The last Column is 2” wide for a 2” Binder.

Evoqua Water Technologies, LLC
558 Clark Road
Tewksbury, MA 01876
(978)-863-4603

Chemical Induction System

EQUIPMENT SUBMITTAL

Rehab and Replacement Ph 1

Hilo, HI

Evoqua Water Technologies, LLC
1619831

Evoqua Water Technologies, LLC
558 Clark Road
Tewksbury, MA 01876
(978)-863-4603

Chemical Induction System

EQUIPMENT SUBMITTAL

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Chemical Induction System

EQUIPMENT SUBMITTAL

Rehab and Replacement Ph 1

Hilo, HI

Evoqua Water Technologies, LLC
1619831

Chemical Induction System

Evoqua Water Technologies Project #1619831

Rehab and Replacement Ph 1,
Hilo, HI

Equipment
Submittal

Submitted To:
Customer: Nan Inc.
Job/PO #: 1619831 / #24077-00013

Prepared By:
Evoqua Water Technologies
558 Clark Road
Tewksbury, MA 01876
987.863.4603
Project Manager – Bruce Sisoukraj
Date: Thursday, September 11, 2025

TABLE OF CONTENTS

Tab 1

SECTION 1.0

STATEMENT OF CONFIDENTIALITY

This document and all information contained herein are the property of Evoqua and/or its affiliates. The design concepts and information contained herein are proprietary to Evoqua and are submitted in confidence. They are not transferable and must be used only for the purpose for which the document is expressly loaned. They must not be disclosed, reproduced, loaned or used in any other manner without the express written consent of Evoqua. In no event shall they be used in any manner detrimental to the interest of Evoqua. All patent rights are reserved. Upon the demand of Evoqua, this document, along with all copies and extracts, and all related notes and analyses, must be returned to Evoqua or destroyed, as instructed by Evoqua. Acceptance of this delivery of this document constitutes agreement to these terms and conditions.

Tab 2

SECTION 2.0

<u>SUBMITTAL APPROVAL FORM</u>

JOB NAME: Rehab and Replacement Ph 1
JOB NUMBER: 1619831
TYPE OF EQUIPMENT: Water Champ Induction System

CHECKED BY: Bruce Sisoukraj DATE: 9/11/2025

We propose to furnish equipment as herein described for the above named job.

This engineering data is submitted as generally complying with the intent of the plans and specifications for this project. However, in the interest of standardization of components and fabrication methods, there are necessarily some exceptions taken. The customer should carefully examine the data submitted herein, reviewing especially for correct dimensions and locations of connections with piping and wiring furnished under other sections of the contract.

Evoqua proposes to furnish only the equipment described in this submittal and provide such only under terms of the sales agreement. Evoqua assumes no responsibility for the equipment not expressly mentioned. Approval of this submittal data constitutes acceptance, becomes binding upon both parties to the agreement, and constitutes a **"NOTICE TO PROCEED WITH MANUFACTURE". APPROVED, SUBJECT TO CORRECTIONS AS NOTED:**

DATE: ____

ENGINEER: ____

BY: ____

Note:
Evoqua Water Technologies, LLC will accept no back charges incurred by any party, for any reason, which have not been specifically authorized in writing by an officer of the company with a cost approved prior to the work being done.

REVIEW AND APPROVAL

Once you have reviewed the submittal documentation, please return/email submittal package with the engineer's stamp of approval and clearly identify any remarks or concerns that you may have. If, at any time during the review, you have any questions or need additional information, please feel free to contact us.

SHIPPING INSTRUCTIONS

Please confirm the address of the site, and provide the telephone number, contact person and any tagging information and/or shipment instructions. Evoqua will look to default to contract address unless revised.

RECEIVING EQUIPMENT

Upon receipt of the shipment, please check the packing lists and Bill of Lading to make sure all items have been received. Should any discrepancy appear, please contact us. Should any part of the shipment arrive damaged or missing, you should request a carrier inspection, file a damage and loss claim report with the carrier, and send a copy to Evoqua. Purchaser is accountable to keep equipment and/or material in a safe, dry place, protected from the elements, vandals and/or thieves. (Please refer to the Shipping, Handling and Storage section of the submittal.) Items found to be missing, lost or damaged, and expenses incurred because of lost time by field personnel as a result of lost or damaged parts, will be at Purchaser's expense.

STARTUP AND TECHNICAL DIRECTION

If start-up service and instructional training are included, please provide at least (3)-Three weeks advance notice prior to actual startup and instructional training. Contact the Evoqua Water Technologies, LLC Field Service Technician at +1 (630) 234-3576 or by email at ronny.parayno@xylem.com.

Tab 3

SECTION 3.0
SCOPE OF SUPPLY

	Chemical Induction Mixer
1	Water Champ SWC2F Chemical Induction Unit, complete with:
	316SS Franklin motor, 4" dia. submersible type, 2HP
	titanium construction for all primary wetted components,
	propellor, 25' power cable, 460V/380V, 3PH, 60/50Hz
1	ALF assembly for 4" motor, with check valve and 3/4" male Camlock connection
	for alternate feed point of chemical to induction unit mixing zone.
1	Hose Assembly, 3/4" x 25 ft., w/ Back Pressure valve
	Guide Rail
1	Guide Rail, 316SS, 6-10 ft
1	Base, Guide Rail Waterchamp
1	Universal Pivot Bracket, fits 4" Motor
	Control Panel
1	System Control Panel, 2 HP model, 460V, 3PH, 60Hz input,
	wall-mount, NEMA 4X fiberglass enclosure with viewing window,
	fused disconnect switch, H-O-A switch for local or remote start-stop control,
	motor protection device for detecting and displaying alarm conditions,
	automatic restart for temporary fault conditions, hour meter,
	run and stop lights, dry contacts for alarm and run/stop status
1	Custom sunshade for outdoor installation
	Spare Parts
1	Spare hose 3/4" x 25 ft., w/ Camlock fittings
2	Spare seals for 3/4" Camlock fittings, Viton (2 per hose)
1	Spare 4" Vacuum Enhancer for 4" Motors
2	Enhancer set screws (2 per unit)
1	Lab Seal for 4" Vacuum body
1	Siesmic/Anchor Calcs and PE Stamps

SCOPE OF SUPPLY BY OTHERS

This list is provided for your information. It does not include all items required to complete this project. Refer to the submittal drawings for additional clarification.

- Installation of foundation bolts
- Sampling Valve
- Pits and concrete work
- Grout and grouting
- Electrical wire and wiring
- Electrical controls (except where stated herein)
- Unloading, erection and storage of equipment
- Handrails, grating and walkways
- Trash containers
- Shims
- Field paint and painting (if required)
- Locating and mounting of panels, local control stations, ultrasonic sensor/transducers and the channel mounted float switches.
- All conduit, fittings, junction boxes, mounting hardware, anchor bolts and electrical wiring and wire between the equipment and control panels, local control stations, junction boxes, ultrasonic sensor/transducers and any mounted float switches.

ITEMS NOT INCLUDED IN SCOPE

- Mechanical and electrical installation and labor
- Civil work including supply of anchor bolts
- Interconnecting piping
- Interconnecting wiring (unless detailed in scope)
- Valves, fittings, appurtenances not specifically listed in the Scope of Supply
- All taxes, fees, lien waivers, bonds and licenses
- Room ventilation, air conditioning or lighting
- Videotaping (unless a videotape agreement is signed)

Tab 4

SECTION 4.0

CLARIFICATIONS AND EXCEPTIONS

Evoqua Water Technologies, LLC propose to furnish materials, and/or equipment for this project. Any items not shown above as detailed under "SCOPE OF SUPPLY", or other attachments to this proposal are EXCLUDED.

	Section	Part	Description
	NOTICE		The scope of supply and pricing are based on Evoqua|Xylem's standard equipment selection, standard terms of sale and warranty terms. Any variations from these standards may affect this quotation.
	Spec/DWG markups		This proposal shall include the markup/comments provided by the review attached below. Exceptions/clarifications to each comment are listed below as reference. Insufficient application data was provided at time of proposal generation, therefore if additional application data becomes available later, we reserve the right to requote any changes necessary for a working customer solution. Please have engineer fill out our attached blank Application Worksheet to verify that our scope of supply is appropriate.
1	11348 (Add#4)	1.04.C	We take exception to stringy materials requirement. This is not applicable to our design since we do not pass nay process effluent through our mixer body.
2		1.04.E	We take exception to references to VFD as we do not have VFD controls for our Water Champ mixer. We shall provide our standard control panel offering which works with on/off control of the mixer, and motor protection device for diagnostics and equipment safety shutdown. If any other customizations are crucial, we reserve the right to review the requirements in detail and offer requote if this is outside our standard.
3		2.02.H	We take exception to orifice housing since this is not applicable to our mixer design. Our mixer uses an impeller, and not an orifice hub design.
4		2.02.J, L, M; 2.04.A.2.a & 4	We take exception to providing davit crane, base, and lifting cable assembly per customer (Nan, Inc.) request. This lifting equipment shall be provided by others if need, per a separate spec section.
5		2.02.L & O	It is not clear what this "flange mount" and "seal plate" refers to, therefore we take exception to providing these items.
6		2.03.A.3	We take exception to "running dry for an extended period of time without damage". This is not applicable to our externally water-cooled, submersible motor design and contradicts 2.03.A.1-2 requirements. This language is applicable to oil-filled motor designs.
7		2.03.7	We take exception to references to "strain relief" functionality in the power cables. This does not apply to our design. We use a separate lift cable to pull mixer up from channel. Lift cable attaches to the butt end of the mixer to fully support the mixer weight. No weight shall be supported by the power cable.
8		2.03.A.8-9	These 2 statements contradict each other.

9		2.06.A.4	Special tools are not needed for our mixer operation
	15050	2.23	We take exception to source testing after factory testing for our Water Champ control panels. We are not testing the control panels in our manufacturing facility for the mixers themselves. Control panels shall be tested only in the field during startup of equipment by our authorized service engineer. Exception to reference to source testing in our spec section 11348 part 3.02.C
	01783	1.05	We shall provide 2 years of our mechanical warranty per the terms below.
	01756	3.13.O	Evoqua will require customer to sign our media release documentation if videotaping is done on our training sessions.
	17710	1.04.B.2.a.1	We shall propose our standard VCP in this scope of supply. Any additional, drastic design changes may incur additional costs. We reserve the right to requote if need.
		1.04.B.3 & C; 2.06.B	Our control panel enclosure shall have integral wall mounting flanges and does not include legs. We take exception to floor stand, backpanel, or legs for our VCP. Therefore we will not provide seismic calcs for our panel. This can be done by others who will install our VCP. Drawings do not show VCP installation arrangements.

Tab 5

SECTION 5.0

<u>SCHEDULE</u>

Date of Equipment Submittal – **Thursday, September 11, 2025**

Evoqua Water Technologies, LLC is requesting approval of the submittal by **September 26th, 2025** in order to meet the proposal schedule. Once this office receives an approved submittal package and approved Purchase Order, the release of the manufacturing can commence. Please note equipment fabrication and release will remain paused until such time that we received authorization for released an/or equipment submittal approval. We cannot commit to meeting proposal lead if the approve date expressed herein is surpassed.

At this time the expected ship date for the equipment is **8-10 weeks after Complete Equipment Submittal approval.**

Tab 6

SECTION 6.0

SUPPLEMENT RECEIVING, HANDLING, STORAGE

A. RECEIVING

CAREFUL OFF LOADING OF EQUIPMENT FROM DELIVERY TRUCKS IS REQUIRED TO PROTECT GLASS AND PLASTIC COMPONENTS

B. HANDLING

THE SAME CARE SHOULD BE USED TO LOCALLY TRANSPORT EQUIPMENT TO WAREHOUSE AND/OR INSTALLATION AREA

C. STORAGE

DO NOT DROP OR STORE EQUIPMENT WHERE IT MIGHT TIP OVER. PROTECT EQUIPMENT FROM FALLING OBJECTS.

THE EQUIPMENT MANUFACTURED BY EVOQUA IS NOT DESIGNED FOR OUTDOOR SERVICE OR STORAGE. THEREFORE, WE RECOMMEND THAT THE EQUIPMENT SUPPLIED BY US FOR THIS PROJECT, BE STORED IN AN INDOOR AREA. THIS AREA SHOULD BE PROTECTED FROM THE ELEMENTS AND HIGH HUMIDITY. THE TEMPERATURE SHOULD BE BETWEEN 50ºF AND 100ºF. ANY EQUIPMENT DAMAGE AS A RESULT OF IMPROPER STORAGE CAN NOT BE COVERED UNDER OUR STANDARD GUARANTEE. IN ADDITION, IN ANY SHORT OR LONG TERM STORAGE, WE RECOMMEND THAT YOU MAKE ARRANGEMENTS WITH OUR AUTHORIZED SERVICE REPRESENTATIVE TO INSPECT THE EQUIPMENT PRIOR TO START-UP. THIS INSPECTION WILL IDENTIFY ANY REPAIRS OR PARTS REPLACEMENT NECESSARY FOR PROPER AND SAFE OPERATION AT START-UP. PLEASE NOTE THAT AN INSPECTION OF THIS KIND IS NOT PART OF THE CURRENT ORDER AND WOULD BE ADDED LATER UPON YOUR REQUEST.

General Storage Considerations

The Water Champ® "F" Series submersible motor is a water-lubricated design. The filling solution consists of a non-toxic, 50/50 mix of propylene glycol and water. The solution will prevent damage from freezing down to -30°F (-34°C). Repeated freezing and thawing should be avoided to prevent possible loss of solution.

Extended storage of the motors, either with or without the Water Champ® body installed, may also result in a slight loss of the filling solution. This loss occurs mainly at the shaft seal. While it may not be discernible because the rate is extremely slow, it evaporates as fast as it comes out. In time, the loss can be enough to cause possible damage. When the storage temperature does not exceed 100°F, storage time should be limited to two years. Where storage temperatures reach 130°F, storage time should be limited to one year. All motors that are in storage should be checked for full fluid levels prior to placing the Water Champ® unit into service.

Long Term Storage Requirements

These requirements are provided to ensure satisfactory and trouble free operation following an extended storage period. Storage of equipment does not extend Evoqua' standard warranty period.

These procedures are for storage of Submersible Water Champ® units and pertain primarily to the electric motor. The lower end of the Water Champ® unit does not require any special storage conditions. The "induction end" parts should be protected from dirt and debris.

General:

- Store in heated dry area. Temperatures during storage should not exceed 150°F maximum or 50°F minimum and a maximum relative humidity of 60%.
- Area is to be free from any shock or vibration of 2 mils maximum at 60Hz.
- Keep motor covered with plastic cover but do not seal as this traps moisture.
- Store motor off the floor to reduce moisture pick up.

Motors – Monthly:

- Once a month the shaft and propeller should be rotated counterclockwise (if looking at propeller end of unit) 10 times to ensure proper operation of bearings. The motor is equipped with carbon sleeve bearing and does not require grease packing.

Tab 7

SECTION 7.0

WARRANTY STATEMENT

Except as provided above, for goods sold by Seller to Buyer(s) that are used by Buyer for personal, family or household purposes, Seller warrants the goods to Buyer on the terms of Seller’s limited warranty available on Seller’s website. For any other purpose, Seller warrants that the goods sold to Buyer under the Agreement (with the exception of software, membranes, seals, gaskets, elastomer materials, coatings and other "wear parts” or consumables all of which are not warranted except as otherwise provided in the Proposal) will be (i) built in accordance with the specifications referred to in the Proposal, if such specifications are expressly made a part of the Agreement, and(ii) free from defects in material and workmanship for a period of two (2) years from the date of installation or thirty (30) months from the date of shipment (which date of shipment will not be greater than thirty (30) days after receipt of notice that the goods are ready to ship), whichever occurs first, unless a longer period is specified in the product documentation (the “Warranty”). For services, the warranty period will be three (3) months from the date the services are performed unless otherwise expressly set forth in the Proposal or sales form or order acknowledgment.

Seller will, at its option, either repair or replace any goods which fails to conform with the Warranty; provided, however, that under either option, Seller will not be obligated to remove the defective goods or install the replaced or repaired goods and Buyer will be responsible for all other costs, including service costs, shipping fees and expenses. Buyer’s failure to comply with Seller’s repair or replacement advice will constitute a waiver of Buyer’s rights and render all warranties void. Any parts repaired or replaced by Seller under the Warranty are warranted only for the remaining balance of the warranty period. The Warranty is conditioned on Buyer giving written notice to Seller of any defects in material or workmanship of warranted goods within ten (10) days, or shorter period as dictated by the issue, of the date when any defects are first manifest. Seller will have no warranty obligations to Buyer with respect to any goods or parts of the goods that: (a) have been repaired by third parties other than Seller or without Seller’s written approval; (b) have been subject to misuse, misapplication, neglect, alteration, accident, or physical damage; (c) have been used in a manner contrary to Seller’s instructions for installation, operation and maintenance; (d) have been damaged from ordinary wear and tear, corrosion, or chemical attack; (e) have been damaged due to abnormal conditions, vibration, failure to properly prime, or operation without flow; (f) have been damaged due to a defective power supply or improper electrical protection; (g) have been damaged resulting from the use of accessory equipment not sold by Seller or not approved by Seller in connection with goods supplied by Seller hereunder; or (h) not sold by Seller or its authorized supplier. In any case of goods not manufactured by Seller, there is no warranty from Seller; however, Seller will extend to Buyer any warranty received from Seller’s supplier of such goods.

THE FOREGOING WARRANTY IS EXCLUSIVE AND IN LIEU OF ANY AND ALL OTHER EXPRESS OR IMPLIED WARRANTIES, GUARANTEES, CONDITIONS OR TERMS OF WHATEVER NATURE RELATING TO THE GOODS PROVIDED HEREUNDER, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY EXPRESSLY DISCLAIMED AND EXCLUDED. BUYER’S EXCLUSIVE REMEDY AND SELLER’S AGGREGATE LIABILITY FOR BREACH OF ANY OF THE FOREGOING WARRANTIES ARE LIMITED TO REPAIRING OR REPLACING THE GOODS AND WILL IN ALL CASES BE LIMITED TO THE AMOUNT PAID BY THE BUYER HEREUNDER.

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WATER TECHNOLOGIES

Tab 8

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

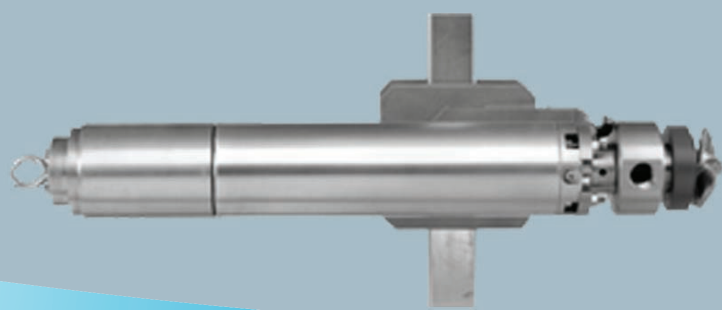
DATE RECEIVED 9/29/2025
SPECIFICATION SECTION # 11348
SPECIFICATION Chemical Mixers
PARAGRAPH 1.04, 1.05, 2.00
DRAWING Control Panel, Pivot bracket , and Water Champ
SUBCONTRACTOR N/A
SUPPLIER EVOQUA
MANUFACTURER N/A

CERTIFIED BY CQCM or Designee : Matthew Chun, QC Engineer

SECTION 8.0

EQUIPMENT/COMPONENT INFORMATION

Tab A



WATER CHAMP® CHEMICAL INDUCTION SYSTEM 4” SUBMERSIBLE SERIES MODEL NO. SWC2F

General Description

The Water Champ® Submersible Chemical Induction System is an innovative device designed for the application of a variety of chemicals used for the treatment of potable water and wastewater. The unique feature of the Water Champ® system is its ability to provide instantaneous mixing and diffusion.

The “F” Series submersible offers the highest quality of design and construction of any submersible chemical induction unit. The hermetically sealed 316 SS motor offers the highest level of durability and performance required for chemical feed applications. Most wetted materials are constructed from Grade 2 Titanium (unalloyed) and are designed for use with most treatment chemicals. The innovative mounting is configured for mounting in open-channel applications and can be easily retrofitted to existing basins and tanks.

Features

- Titanium wetted parts*
- Rugged construction
- Heavy duty bearing design

Key Benefits

- Instantaneous diffusion and mixing
- Energy and chemical savings
- Maximum chemical concentration with no off-gassing
- Warranty - 1 year from start-up or 18 months after delivery

SPECIFICATIONS

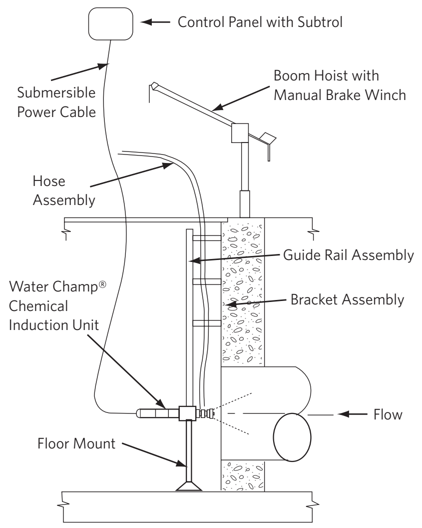
Model No.	KW (HP)	Max. Gas Induction CL^2 kg/d (ppd)	Max. Liquid Induction lpm US gpm	Max. Vacuum kPa (in Hg)
SWC2F	1.5 (2)	226.8 (500**)	56.8 (15**)	.61 (18)

* Custom materials quoted upon request.

** Feed rates are application dependent (consult factory).

MATERIALS OF CONSTRUCTION

Part	Material
Vacuum Chamber	Titanium
Vacuum Port	Titanium
Vacuum Enhancer	Non-Metallic
Propeller	Titanium
Propeller Bolt	Titanium
Shaft	Titanium
Mechanical Seals	Carbon / Ceramic
Motor	316 Stainless Steel
Power Cable	TC Type (25 ft standard)
Hardware	316 Stainless Steel



TYPICAL WATER CHAMP® SYSTEM HORIZONTAL INSTALLATION

MOTOR PERFORMANCE DATA

Model	KW (HP)	Volts	Phase	Hertz	RPM	Insulation	S.F.	FLA Rated Input Amps	SFA Max. Input Amps	Locked Rotor Amps	Power Factor %			Efficiency %		
											S.F.	F.L.	3/4	S.F.	F.L.	3/4
SWC2F	1.5 (2)	200	3	60	3450	Class B	1.25	7.7	9.3	53.6	84.4	79	71.2	69.5	69.5	67.4
SWC2F	1.5 (2)	230	3	60	3450	Class B	1.25	6.7	8.1	46.6	84.4	79	71.2	69.5	69.5	67.4
SWC2F	1.5 (2)	230	1	60	3450	Class B	1.25	Y 10 B 9.3 R 2.6	Y 13.2 B 11.9 R 2.6	51.0	93.1	90.5	86.7	70	71	68.8
SWC2F	1.5 (2)	460	3	60	3450	Class B	1.25	3.4	4.1	23.3	84.4	79	71.2	69.5	69.5	67.4
SWC2F	1.5 (2)	575	3	60	3450	Class B	1.25	2.7	3.2	18.6	84.4	79	71.2	69.5	69.5	67.4

EVOQUA WATER TECHNOLOGIES

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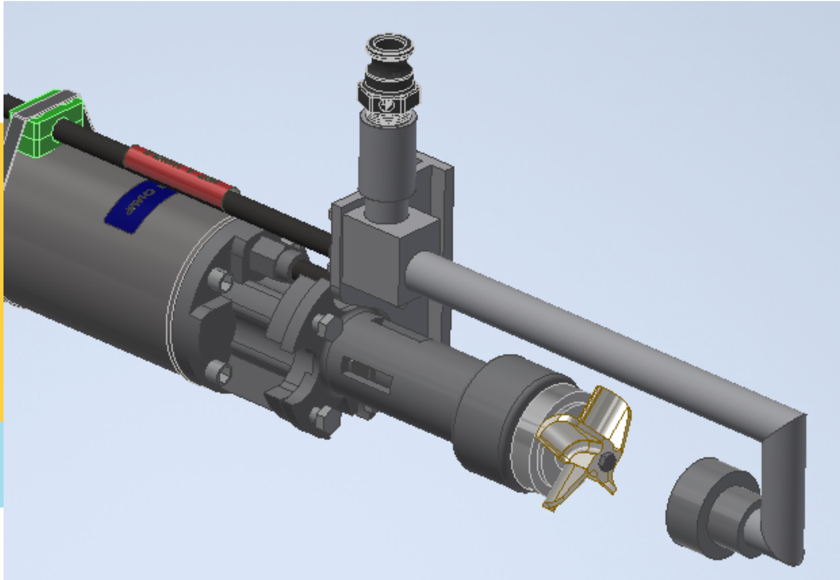
Tab B

EVOQUA
WATER TECHNOLOGIES



*FX Water Champ Shown

WATER CHAMP ALTERNATE LIQUID FEED

General Description

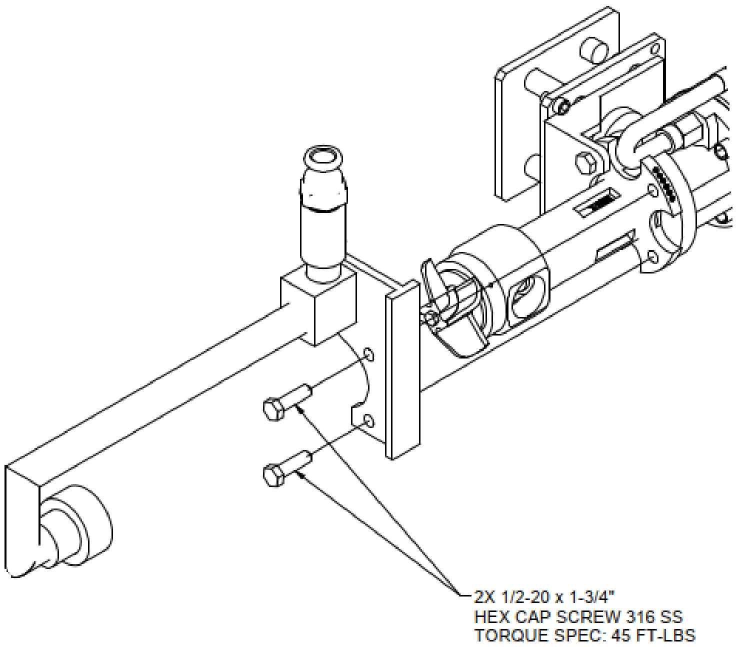
The Water Champ® Submersible Chemical Induction System is an innovative device designed for the application of a variety of chemicals used for the treatment of potable water and wastewater. Some applications may require the use of solutions that will scale and to prevent scaling within the Water Champ the use of the Alternate Liquid Feed (ALF) is recommended.

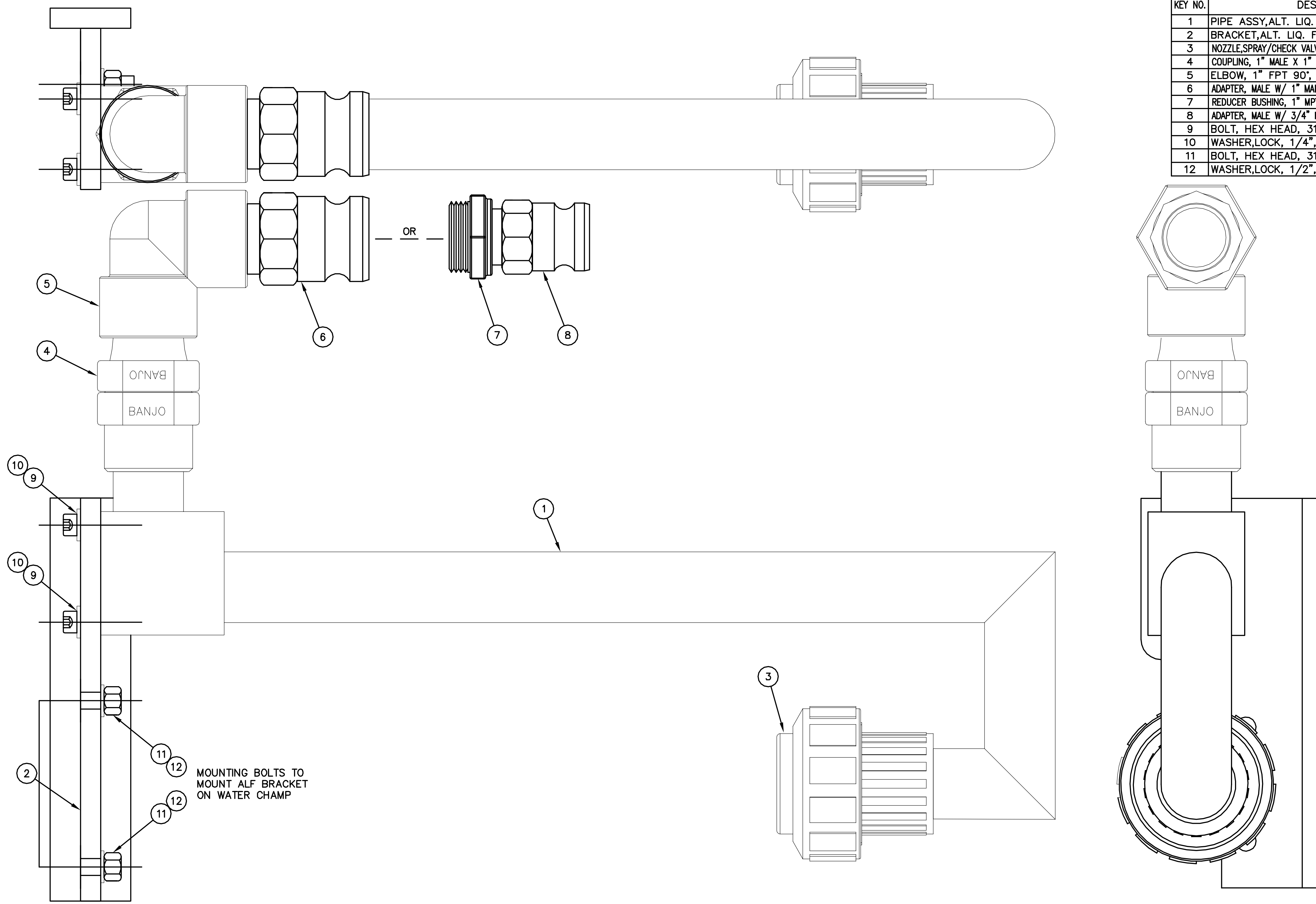
Features

- Titanium wetted parts
- Rugged construction
- 1" Hose Connection

Key Benefits

- Prevents internal scaling
- Easy Bolt-On optional accessory
- No added moving parts
- Added Redundancy





BILL OF MATERIALS

KEY NO.	DESCRIPTION	PART No.	QTY
1	PIPE ASSY,ALT. LIQ. FEED 1"	W3T110093	1
2	BRACKET,ALT. LIQ. FEED 1" FOR 4" MOTOR	W3T193401	1
3	NOZZLE,SPRAY/CHECK VALVE	W2T108028	1
4	COUPLING, 1" MALE X 1" FEMALE SWIVEL, POLYPROPYLENE	W2T1689	1
5	ELBOW, 1" FPT 90°, POLYPROPYLENE	W2T521996	1
6	ADAPTER, MALE W/ 1" MALE THREAD, POLYPROPYLENE	W2T1807	1
7	REDUCER BUSHING, 1" MPT X 3/4" FPT, POLYPROPYLENE	W2T521997	1
8	ADAPTER, MALE W/ 3/4" MALE THREAD, POLYPROPYLENE	W2T265228	1
9	BOLT, HEX HEAD, 316SS,1/4-20 X 3/4"	W2T3451	4
10	WASHER,LOCK, 1/4", 316SS	W2T3701	4
11	BOLT, HEX HEAD, 316SS,1/2-20 X 1-3/4"	W2T5612	2
12	WASHER,LOCK, 1/2", 316SS	W2T2865	2

THIRD ANGLE PROJECTION

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES AND INCLUDE THICKNESS OF COATINGS.
TOLERANCES ON: ALL SURFACES 125
1-PLACE DECIMALS ±.03
2-PLACE DECIMALS ±.015
3-PLACE DECIMALS ±.005
FRACTIONS ±1/8"
ANGLES ±1°
DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH ANSI Y14.5M

REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN
0	ISSUE FOR MANUFACTURING	12/08/15	WJC	WJC	RS	–

DESIGNER WJC	DATE 06/29/15	
CHECKER WJC	DATE 06/29/15	
ENGINEER WJC	DATE 06/29/15	
MANAGER RS	DATE 06/29/15	
FILE: PS-7023_W3T193397_0		
SCALE: 1=1		

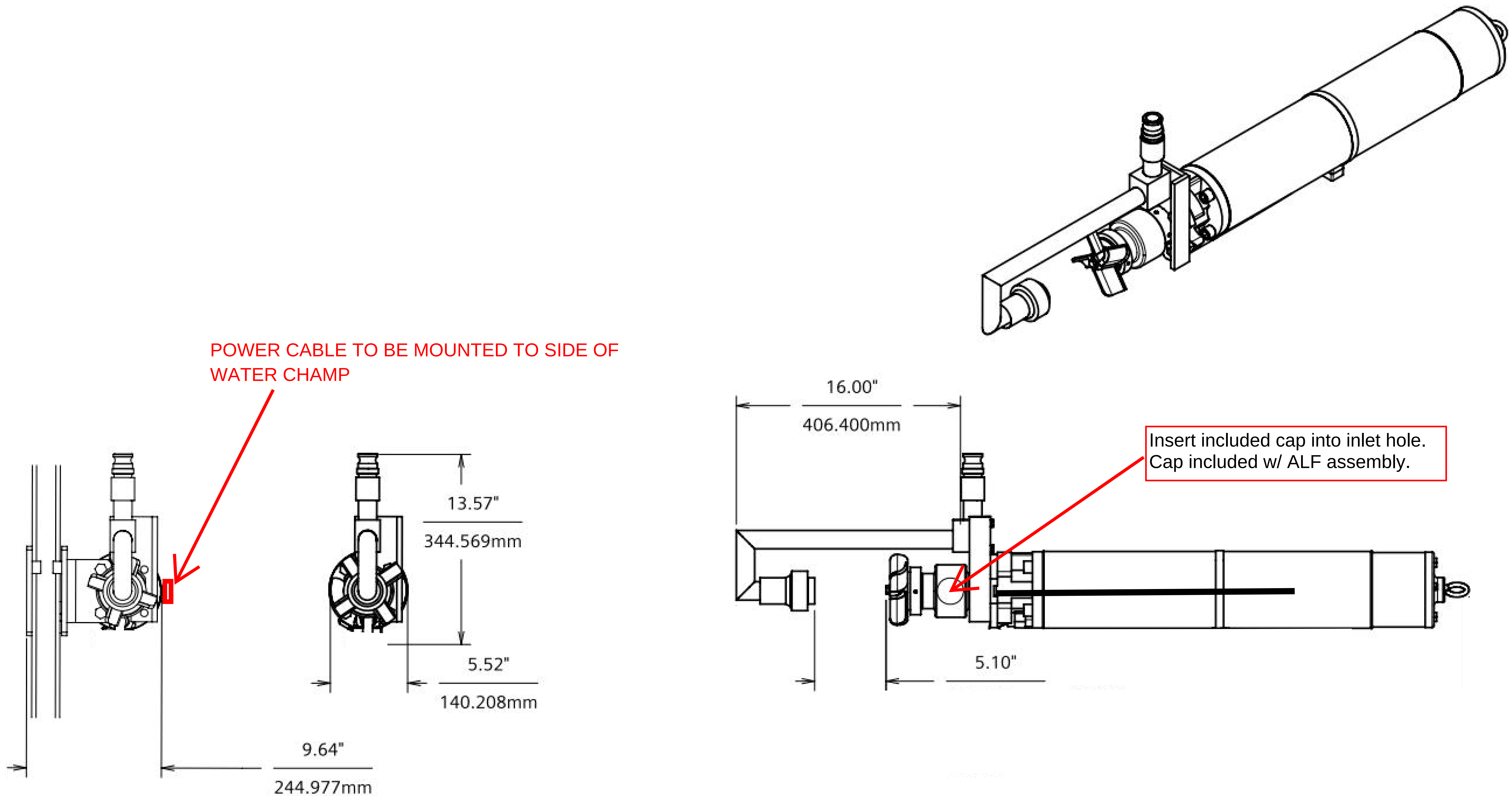
TITLE ASSEMBLY DRAWING
1" ALTERNATE LIQUID FEED (ALF) WITH 3/4" OR 1" HOSE FOR 4" WATER CHAMP MOTOR

CLIENT

EVOQUA WATER TECHNOLOGIES
Water Technologies
Colorado Springs,
CO 719-550-2100

PROJECT	SAP NUMBER	DRAWING	SHEET	REV
WC	W3T193397	PS-7023	1 OF 1	0

BORDER: 22X34D PRINT NOT TO SCALE INTL REF:



REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON DECIMALS/FRACTIONS

.X	±0.1
.XX	±0.01
.XXX	±0.003
ANGLES	±1.0°-0°
SURFACES	125√

DESIGNER steven.jones	DATE 04/15/2008
CHECKER	DATE 01/01/2099
ENGINEER	DATE 01/01/2099
MANAGER	DATE 01/01/2099
FILE: ALF-ON-6-001381_0	
SCALE: 1:8	

TITLE ALT. LIQ. FEED REPRESENTATION

ALT. LIQ. FEED REPRESENTATION

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EVOQUA WATER TECHNOLOGIES

Water Technologies
Colorado Springs, CO
719-550-2100

PROJECT	CODE	DRAWING	SHEET	REV
	ALF-ON-6	ALF-ON-6-0013811	OF 1	0

Tab C

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WATER TECHNOLOGIES

WATER CHAMP® SYSTEM CONTROL PANEL

General Description

The Water Champ® System Control Panel is specifically designed to offer ultimate protection for your submersible Water Champ Chemical Induction System. The Control Panel includes the new Motor Protection Device (MPD).

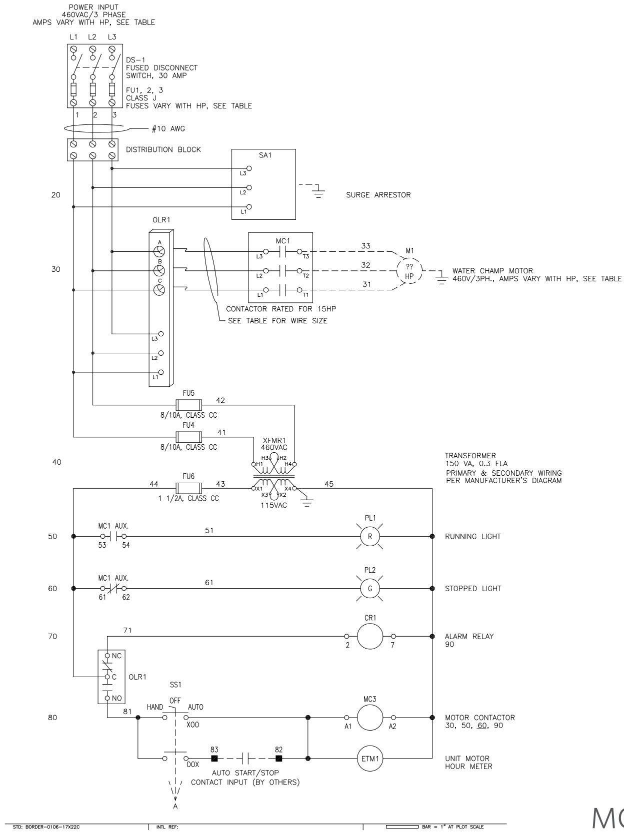
Computer technology is applied to provide a unique system of detecting overloads, underloads, rapid cycling, ground faults, temperature, current and voltage unbalance. The Motor Protection Device will turn off the Water Champ system should any of these faults occur and provide a visual display of the fault condition. In addition, the Motor Protection Device offers automatic restart when the problem is temporary, or can signal an alarm or backup system if it is constant.

The Water Champ Control Panel is housed in a NEMA 4X corrosion resistant enclosure with a viewing window for all operator usable functions and diagnostics.

The easy access control panel allows the user to adjust the reset time and provides real-time voltage and current status to ensure peak performance.

Key Benefits

- Control start/stop of Water Champ® System (locally and remotely)
- Phase reversal protection
- Three phase disconnect with lockout capability
- Surge arrestor which exceeds ANSI / IEEE standard C62.11



MOTOR PROTECTION DEVICE SYSTEM

Motor Protection Device: How It Works

A microprocessor is the main element of the Motor Protection Device system. It continuously monitors the motor's operating condition. The Motor Protection Device will immediately shut down the motor if a fault condition occurs. To assure that conditions are not temporary, the Motor Protection Device will attempt to restart the motor. If the alarm condition still exists after three restart attempts, a manual restart is required.

The Motor Protection Device protects the Water Champ® System from a variety of application conditions by monitoring some very simple inputs. The Motor Protection Device monitors motor current draw and trips if the amps are either too high, too low, or unbalanced (both overload and underload trip points are field adjustable). In the event of a fault, the Motor Protection Device forces the motor to remain disengaged at least three minutes between run cycles.

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WATER TECHNOLOGIES

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Tab D

EVOQUA
WATER TECHNOLOGIES

WATER CHAMP® CHEMICAL INDUCTION UNIT MOTOR PROTECTION DEVICE

The MPD is a fully-programmable electronic overload relay. An alphanumeric LED display provides programming and diagnostic information. Seventeen parameters can be programmed in the MPD.

1. Low Voltage Set Point
2. High Voltage Set Point
3. Voltage Unbalance Set Point
4. CT Size/Loop Setting
5. Overcurrent Trip Point
6. Undercurrent Trip Point
7. Current Unbalance Trip Point
8. Trip Class (5,10,15,20,30)
9. Rapid Cycle Timer (RD1)
10. Overload Restart Delay (RD2)
11. Underload Restart Delay (RD3)
12. No. of restarts after an overload (Manual or Automatic)
13. RS485 Address
14. No. of restarts after an underload fault
15. Underload Trip Delay
16. Ground Fault Trip Point
17. Temperature Monitoring Sensor

Features

- Recordable voltage, current, last 4 faults, KWh usage, and power factor is available when using communications package
- Digitally programmable for precise customizing.
- Seventeen set points can be programmed for maximum protection
- Last fault memory provides instant troubleshooting diagnostics
- UL® and CE listed as an overload relay
- RS-485 communication port for use with computerized systems using Modbus protocol

Key Benefits

- Overload or underload motor protection
- Over or under voltage change protection
- Protects against single-phasing or unbalanced voltage/current
- Built-in time prevents rapid cycling

SPECIFICATIONS

Electrical	
Input voltage	200-480 VAC, 3-ph (Standard) 500-600 VAC for model MPD-575
Frequency	50-60 Hz
Motor full load amp range	2-25A, 3 ph (loops required) 25-90A, 3 ph (direct) 80-800A, 3 ph (external CT's)
Power consumption	10W (maximum)
Output contact rating SPDT (Form C)	Pilot duty rating: 480 VA @ 240 VAC General purpose: 10 A @ 240 VAC
Expected Life	
Mechanical	1 x 10^6 operations
Electrical	1 x 10^5 operations at rated load
Accuracy at 77° F (25°C)	
Voltage	± 1%
Current	± 3% (< 100A direct)
GF Current	± 15%
Timing	5% ± 1% second
Repeatability	
Voltage	± 0.5% of nominal voltage
Current	± 1% (<100A direct)
Trip times (those not shown have user selectable trip times.)	
Ground Fault Trip Time	Trip time
101%-200% of Setpoint	8 seconds 41 second
201%-300% of Setpoint	4 seconds 41 second
301%-400% of Setpoint	3 seconds 41 second
401% or Greater	2 seconds 41 second
Current Unbalance Trip Times	
% Over Setpoint	Trip time
1%	30 seconds
2%	15 seconds
3%	10 seconds
4%	7.5 seconds
5%	6 seconds
6%	5 seconds
10%	3 seconds
15%	2 seconds

Safety Marks	
UL® Standards	UL508, UL1053
CE	IEC 60947-1, IEC 60947-5-1
Standards Passed	
Electrostatic Discharge (ESD) Radio Frequency Immunity (RFI), Conducted Radio Frequency Immunity (RFI), Radiated	IEC 1000-4-2, Level 3, 6kV contact, 8kV air IEC 1000-4-6, Level 3 10V/m IEC 1000-4-3, level 3 10V/m
Fast transient burst	IEC 1000-4-4, Level 3, 3.5 kV input power
Surge	
IEC	1000-4-5 Level 3, 2kV line-to-line; Level 4, 4kV line-toground
ANSI/IEEE	
Hi-potential test	Meets UL508 (2x rated V +1000V for 1 minute)
Mechanical	
Dimensions	3.0"H x 5.1"D x 3.6"W
Terminal torque	7 in lbs.
Enclosure material	Polycarbonate
Weight	1.2 lbs
Maximum conductor size through MPD	0.65" with insulation
Environmental	
Temperature range	Ambient Operating: -4º - 158ºF (-20º -70º C)
Class of protection	IP20, NEMA 1
Relativity humidity	10-95%, non-condensing per IEC 68-2-3

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Tab E

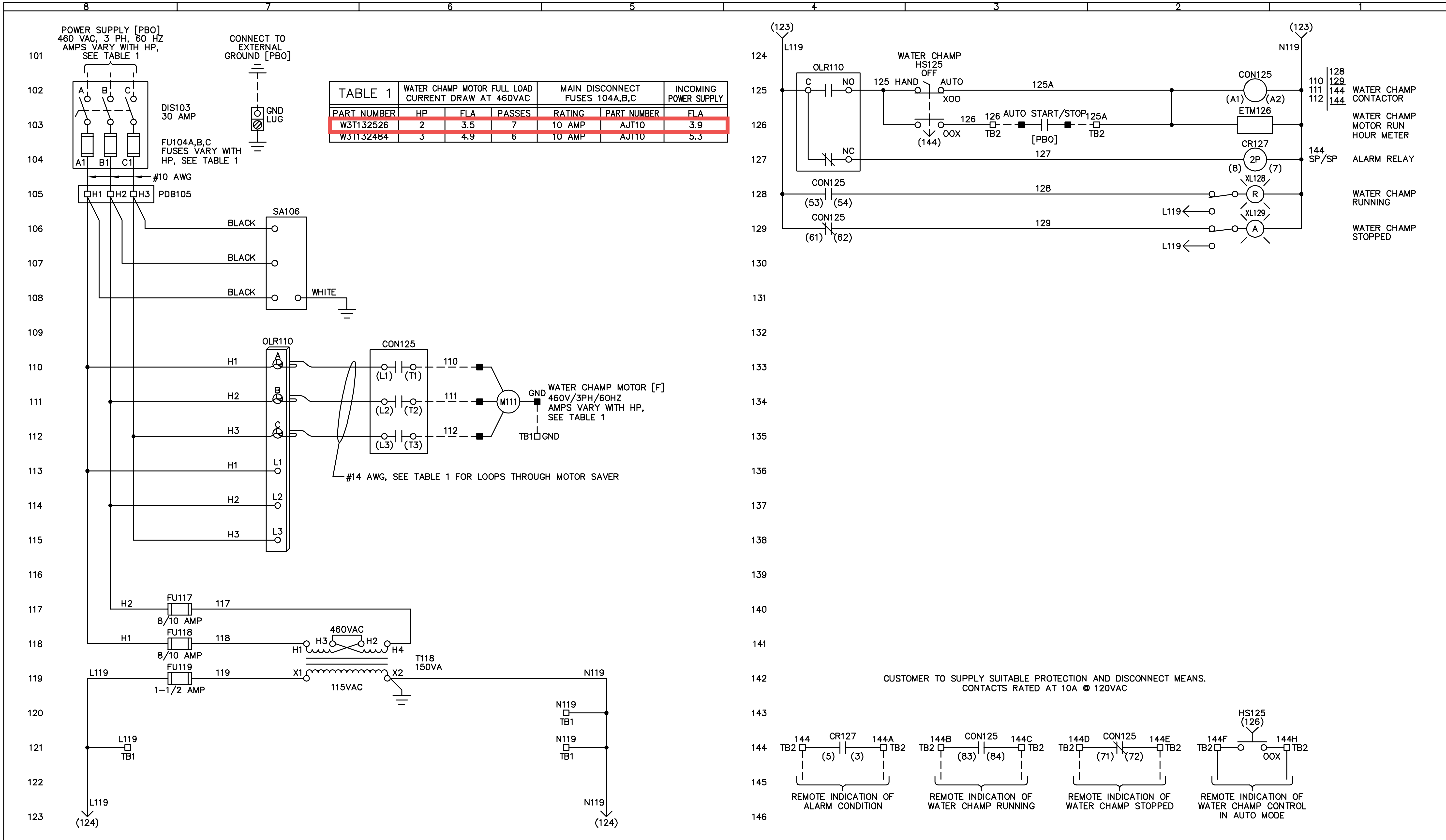


TABLE 1

PART NUMBER	WATER CHAMP MOTOR FULL LOAD CURRENT DRAW AT 460VAC			MAIN DISCONNECT FUSES 104A,B,C		INCOMING POWER SUPPLY
	HP	FLA	PASSES	RATING	PART NUMBER	FLA
W3T132526	2	3.5	7	10 AMP	AJT10	3.9
W3T132484	3	4.9	6	10 AMP	AJT10	5.3

THIRD ANGLE PROJECTION

CONTROL PANEL
UL COMPLIANCE: MATERIALS AND ASSEMBLIES MANUFACTURED WITHIN SCOPE OF UNDERWRITERS LABORATORIES SHALL CONFORM TO UL STANDARDS AND HAVE AN APPLIED UL LISTING MARK.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES AND INCLUDE THICKNESS OF COATINGS.
TOLERANCES ON: ALL SURFACES 125
1-PLACE DECIMALS ±.03
2-PLACE DECIMALS ±.015
3-PLACE DECIMALS ±.005
FRACTIONS ±1/8"
ANGLES ±1°
DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH ANSI Y14.5M

REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN
4	ISSUE FOR RECORD	03/07/17	RJS	RJS	RJS	–

DESIGNER	RJS	03/07/17
CHECKER	RJS	03/07/17
ENGINEER	RJS	05/19/10
MANAGER		

FILE: PS-7006_4
SCALE: 1=1

TITLE: WATER CHAMP, MODEL SWCF 2 & 3HP CHEMICAL INDUCTION SYSTEM CONTROL PANEL

CLIENT

EVOQUA WATER TECHNOLOGIES
Water Technologies
Colorado Springs,
CO 719-550-2100

PROJECT	SAP NUMBER	DRAWING	SHEET	REV
WC	SEE TABLE 1	PS-7006	1 OF 5	4

BORDER: 22X34D PRINT NOT TO SCALE INTL REF:

SWITCH CONFIGURATIONS

ZS	NORMALLY OPEN CLOSE ON INCREASE
ZS	NORMALLY OPEN HELD CLOSED
ZS	NORMALLY CLOSED OPEN ON INCREASE
ZS	NORMALLY CLOSED HELD OPEN
LS	FLOAT SWITCH
PS	PRESSURE SWITCH
PS	DIFFERENTIAL PRESSURE SWITCH
TR	ON DELAY TIMED TO CLOSE
TR	OFF DELAY TIMED TO OPEN
CR NNN	NORMALLY OPEN CONTACT OF A DEVICE LOCATED AT LEVEL NNN
CR NNN	NORMALLY CLOSED CONTACT OF A DEVICE LOCATED AT LEVEL NNN
FS	FLOW SWITCH
TS	TEMPERATURE SWITCH
FTS	FOOT SWITCH
	MUSHROOMHEAD PUSHBUTTON
TR	ON DELAY TIMED TO OPEN
TR	OFF DELAY TIMED TO CLOSE

NOTE: SWITCH POSITION IS TO REFLECT THE NORMAL/DE-ACTIVATED STATE

TERMINAL AND WIRING SYMBOLOGY

NNN	TERMINAL IN EVOQUA PANEL
NNN	TERMINAL IN A DIFFERENT PANEL, FIELD DEVICE, OR MOTOR CONTROL CENTER
	WIRE WITH SHIELD
	PLUG IN CONNECTOR
	PANEL/FACTORY WIRING
	FUTURE WIRING
	FIELD WIRING
	WIRES ARE CONNECTED
	WIRES ARE NOT CONNECTED
(NNN)	WIRE IS CONTINUED TO LADDER NNN
(NNN)	WIRE IS CONTINUED FROM LADDER NNN
[PBO]	INDICATES THE DEVICE SHOWN IS PROVIDED BY OTHER THAN EVOQUA
[F]	INDICATES A FIELD MOUNTED DEVICE

PILOT DEVICE CONFIGURATIONS

TYPICAL SELECTOR SWITCH

501 HS XXX OFF HAND AUTO XOO
502 OOX (601)

- HS XXX: REFERENCES SWITCH OPERATOR LOOP NUMBER OR LADDER POSITION
- XOO: CONTACT CLOSED ONLY IN THE EXTREME LEFT HAND POSITION
- OOX: CONTACT CLOSED ONLY IN THE EXTREME RIGHT HAND POSITION
- (601): INDICATES THAT SWITCH XXX HAS ANOTHER CONTACT AT LEVEL 601

601 HS XXX 502 OXX

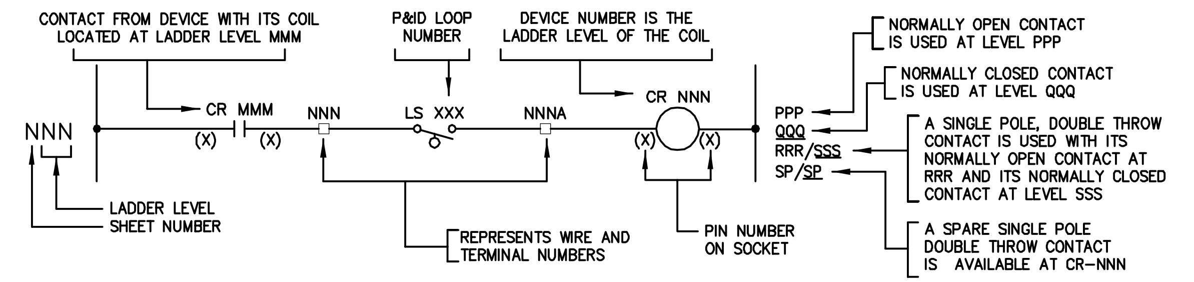
- HS XXX: REFERENCES SWITCH OPERATOR LOOP NUMBER OR LADDER POSITION
- OXX: CONTACT CLOSED IN RIGHT TWO POSITIONS ONLY

200 HS XXX — TYPICAL FOR MOMENTARY PUSH BUTTON SWITCH

300 XL XXX Y — PILOT LIGHT – PUSH–TO–TEST
Y = COLOR: R = RED, G = GREEN, W = WHITE, A = AMBER, L = BLUE

400 XL XXX Y — PILOT LIGHT
Y = COLOR: R = RED, G = GREEN, W = WHITE, A = AMBER, L = BLUE

LADDER NUMBERING CONVENTION AND COMPONENT IDENTIFICATION



DEVICE PREFIX INDICATES GENERAL DEVICE TYPE AS:

ALR = ALARM HORN, BELL OR BUZZER
CB = CIRCUIT BREAKER
CR = CONTROL RELAY
DIS = DISCONNECT SWITCH
FE = FLOW ELEMENT
FI = FLOW INDICATOR
FS = FLOW SWITCH
FT = FLOW TRANSMITTER
FU = FUSE
HS = SELECTOR/HAND SWITCH
HTR = HEATER
LA = LIGHTNING ARRESTOR
LE = LEVEL ELEMENT
LF = LINE FILTER
LI = LEVEL INDICATOR
LIC = LEVEL INDICATING CONTROLLER
LS = LEVEL SWITCH
LT = LEVEL TRANSMITTER
MS = MOTOR STARTER COIL
PI = PRESSURE INDICATOR (OR GAUGE)
PS = PRESSURE SWITCH
PT = PRESSURE TRANSMITTER
PWR = POWER SUPPLY
T = TRANSFORMER
TR = TIMER
TS = TEMPERATURE SWITCH
XL = PILOT LIGHT
ZS = LIMIT SWITCH

MISCELLANEOUS SYMBOLS

MCBX XXA	MAIN CIRCUIT BREAKER
MCPX XXA	3-PHASE MOTOR CIRCUIT PROTECTOR
	3-PHASE DISCONNECT
CBX	CIRCUIT BREAKER
FUX A	FUSE
FU A	FUSED SWITCH
	CHANGE NOTICE SYMBOL
M	3-PHASE MOTOR
M	SINGLE-PHASE MOTOR
OL	THERMAL ELEMENT ON MOTOR STARTER
H1 H3 H2 H4 X1 X3 X2 X4	TRANSFORMER
SVX	SOLENOID VALVE
D	DIODE
D	ZENER DIODE
	PANEL GROUND
	EARTH GROUND
ISOL	ISOLATED INSTRUMENT GROUND
RE	DUPLEX RECEPTACLE
ALR	ALARM HORN
	POT
HTR X WATTS	HEATER
R X OHMS	RESISTOR
C	CAPACITOR

THIRD ANGLE PROJECTION

CONTROL PANEL
UL COMPLIANCE: MATERIALS AND ASSEMBLIES MANUFACTURED WITHIN SCOPE OF UNDERWRITERS LABORATORIES SHALL CONFORM TO UL STANDARDS AND HAVE AN APPLIED UL LISTING MARK.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES AND INCLUDE THICKNESS OF COATINGS.
TOLERANCES ON: ALL SURFACES 125
1–PLACE DECIMALS ±.03
2–PLACE DECIMALS ±.015
3–PLACE DECIMALS ±.005
FRACTIONS ±1/8"
ANGLES ±1°
DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH ANSI Y14.5M

REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN
3	ISSUE FOR RECORD	12/23/14	WJC	WJC	DC	–

DESIGNER WJC	DATE 05/19/10	
CHECKER WJC	DATE 05/19/10	
ENGINEER WJC	DATE 05/19/10	
MANAGER	DATE	
FILE: PS-7006_3		
SCALE: 1=1		

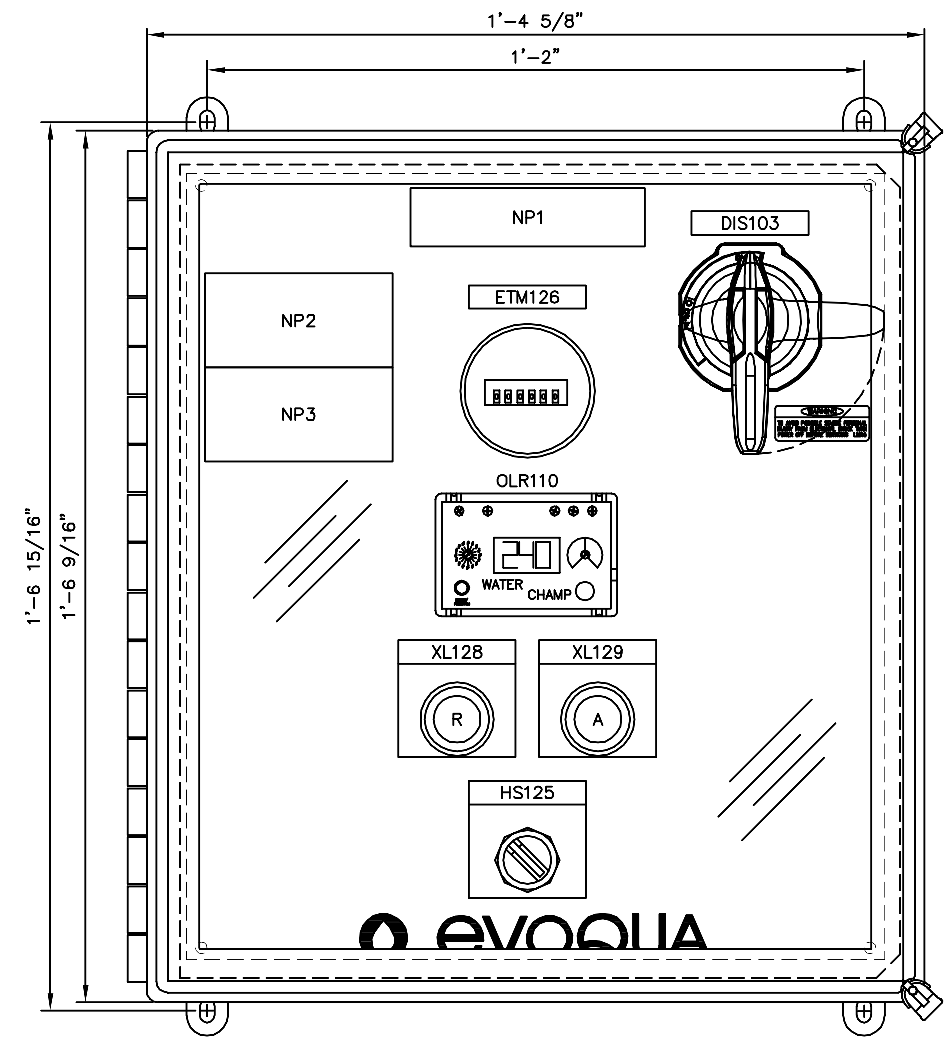
TITLE WATER CHAMP, MODEL SWCF 2 & 3HP CHEMICAL INDUCTION SYSTEM CONTROL PANEL

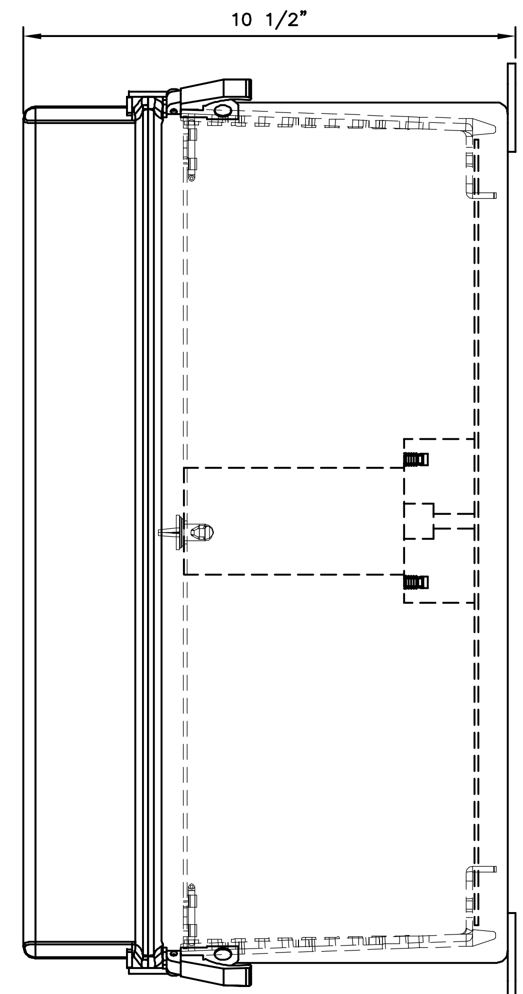
CLIENT

EVOQUA WATER TECHNOLOGIES
Water Technologies
Colorado Springs,
CO 719–550–2100

PROJECT	SAP NUMBER	DRAWING	SHEET	REV
WC	SEE TABLE 1	PS–7006	2 OF 5	3

BORDER: 22X34D PRINT NOT TO SCALE INTL REF:





HOFFMAN NEMA TYPE 4X FIBERGLASS ENCLOSURE W/WINDOW
CATALOG NO. A181610CHQRFGW
HOFFMAN SWING-OUT PANEL
CATALOG NO. A18SPK16C
HOFFMAN STEEL RELAY PANEL
CATALOG NO. A18P16
FINISH: NONE

MASTER NAMEPLATES

LOCATION	INSCRIPTION	STYLE	QTY
NP1	WATER CHAMP LOCAL CONTROL PANEL	STYLE 1	1
NP2	SEE SHEET 5		1
NP3	SEE SHEET 5		1

LEGEND/NAMEPLATES

LOCATION	INSCRIPTION LINE 1 LINE 2 LINE 3 LINE 4 LINE 5	STYLE
DIS103	MAIN DISCONNECT	STYLE 2
ETM126	WATER CHAMP MOTOR RUN HOURS	STYLE 2
HS125	WATER CHAMP CONTROL MODE HAND OFF AUTO	STYLE 3
XL128	WATER CHAMP RUNNING	STYLE 3
XL129	WATER CHAMP STOPPED	STYLE 3

ALL LEGEND PLATES AND NAMEPLATES TO BE PHENOLIC MATERIAL WITH WHITE BACKGROUND AND BLACK LETTERING. LETTER HEIGHT 1/8" HIGH

FOR NAMEPLATE STYLE LAYOUT, SEE SHEET 5 OF THIS DRAWING.

THIRD ANGLE PROJECTION

CONTROL PANEL
UL COMPLIANCE: MATERIALS AND ASSEMBLIES MANUFACTURED WITHIN SCOPE OF UNDERWRITERS LABORATORIES SHALL CONFORM TO UL STANDARDS AND HAVE AN APPLIED UL LISTING MARK.

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FRACTIONS ±1/8"
ANGLES ±1°
DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH ANSI Y14.5M

REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN
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ENGINEER	WJC	DATE	05/19/10
MANAGER		DATE	

FILE: PS-7006_3
SCALE: 0'-6"=1'-0"

TITLE WATER CHAMP, MODEL SWCF 2 & 3HP
CHEMICAL INDUCTION SYSTEM
CONTROL PANEL

CLIENT

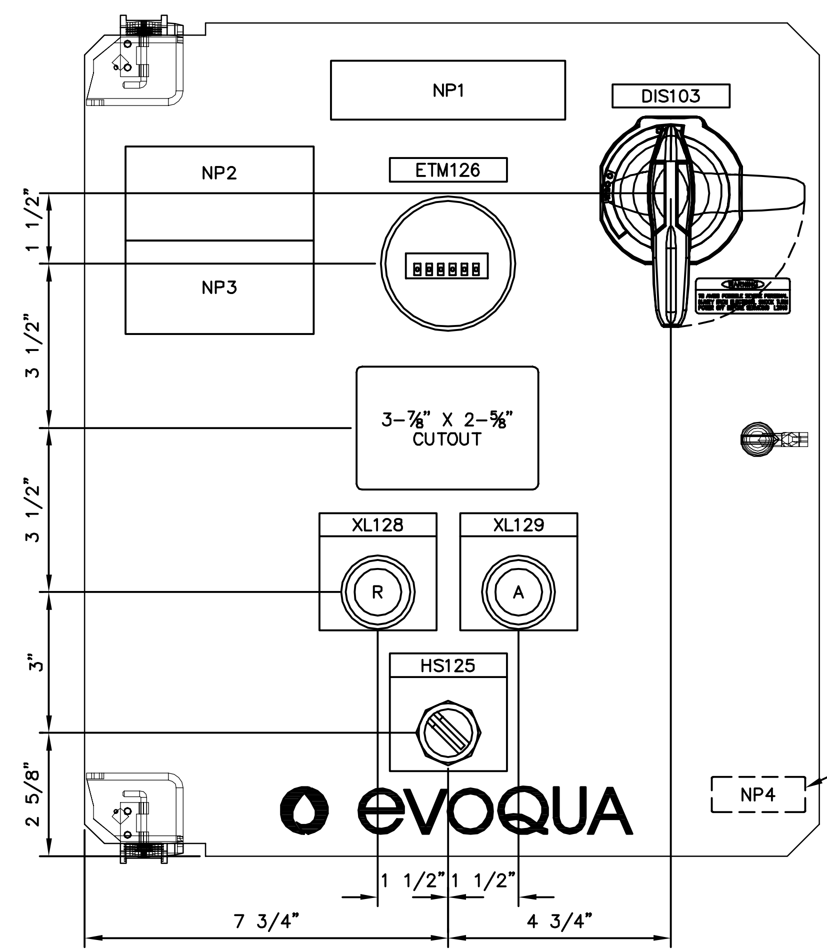
EVOQUA WATER TECHNOLOGIES
Water Technologies
Colorado Springs,
CO 719-550-2100

PROJECT	SAP NUMBER	DRAWING	SHEET	REV
WC	SEE TABLE 1	PS-7006	3 OF 5	3

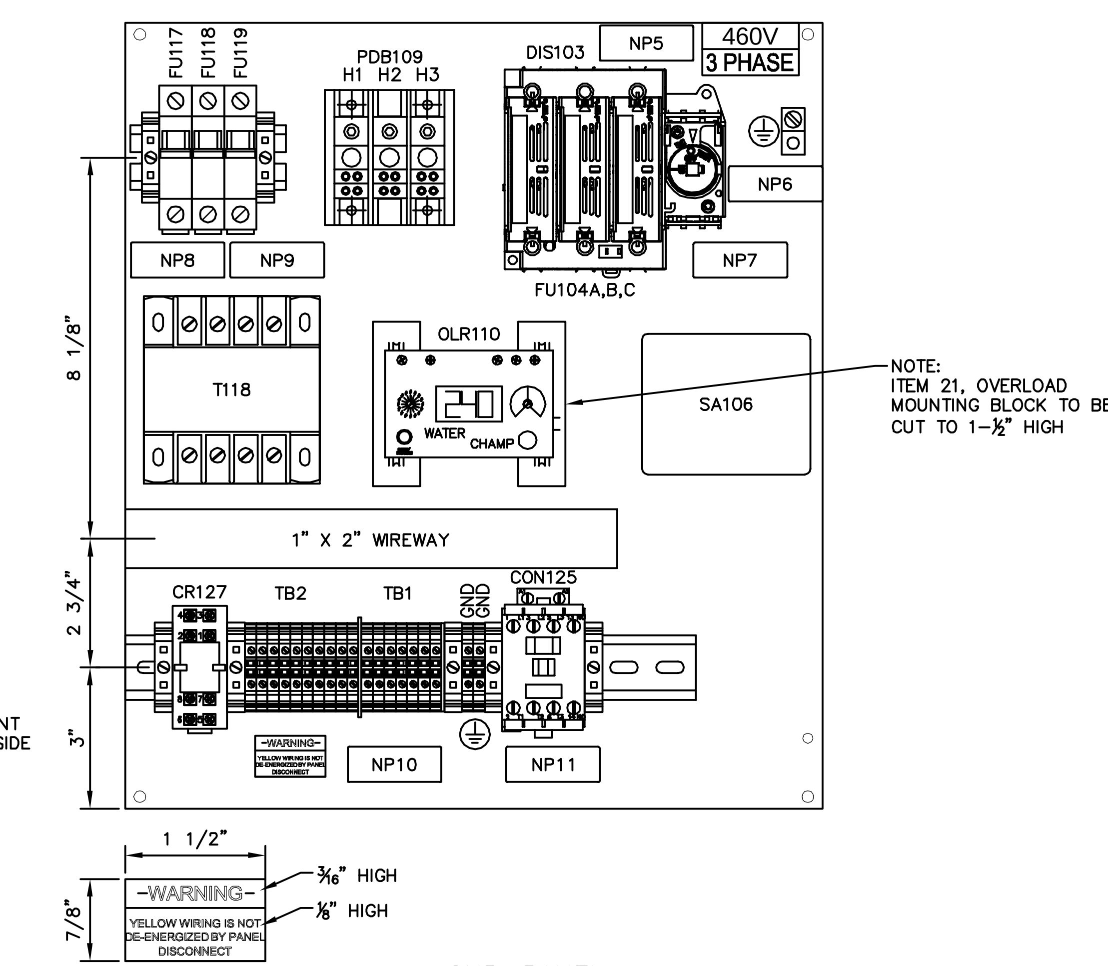
BORDER: 22X34D PRINT NOT TO SCALE INTL REF:

NOTES:
1. HOLES FOR COMPONENTS MOUNTED ON THE ENCLOSURE DOOR AND SUB-PANEL TO BE CLEAN CUT AND FREE OF BURRS.
2. ALL CONTROL PANEL COMPONENTS SHALL BE MOUNTED IN ACCORDANCE WITH UL-508A.
3. IDENTIFY ALL COMPONENTS WITH ENGRAVED PLASTIC NAMEPLATES, WHITE BACKGROUND AND BLACK LETTERING.
4. ALL CONTROL PANEL WIRING SHALL HAVE A TEMPERATURE RATING OF 90°C (194°F) 600V AND SHALL COMPLY WITH THE MOST CURRENT UL-508A GUIDELINE, HOWEVER NO CONDUCTOR MAY BE SUBJECTED TO LOADS GREATER THAN ITS PUBLISHED 75°C (167°F) AMPACITY. WIRE COLORS SHALL CONFORM TO TABLE BELOW.
5. ALL WIRES TO BE IDENTIFIED AT BOTH ENDS WITH WIRE MARKERS. PRINTING FOR WIRE MARKERS AND TERMINAL BLOCK STRIPS MUST BE DONE MECHANICALLY.
6. WIRING SHALL BE ARRANGED NEATLY, CUT TO LENGTH WITH SURPLUS REMOVED. ABRASION PROTECTION MUST BE APPLIED TO WIRING AT HINGE POINTS. NO SPLICES OR WIRE-NUTS ARE PERMITTED; ALL TERMINATIONS MUST BE MADE TO COMPONENT TERMINALS OR TERMINAL BLOCKS.

POWER AND CONTROL CONDUCTORS	
BLACK	UNGROUNDED LINE, LOAD AND CONTROL POWER CONDUCTORS UP TO THE LAST SHORT CIRCUIT/OVERCURRENT DEVICE
WHITE	GROUNDED AC NEUTRAL CONDUCTORS
RED	UNGROUNDED 120VAC CONTROL CIRCUIT CONDUCTORS
BROWN	UNGROUNDED 24VAC CONTROL CIRCUIT CONDUCTORS
GREY	GROUNDED OR UNGROUNDED 24VAC NEUTRAL CONDUCTORS
YELLOW	UNGROUNDED CONTROL CONDUCTORS THAT MAY REMAIN ENERGIZED WHEN THE DISCONNECT IS IN THE "OFF" POSITION
BLUE	UNGROUNDED DC (+) CONTROL CONDUCTORS
BLUE W/WHT TRACER	ANY DC COMMON (−) CONTROL CONDUCTORS
GREEN OR GRN/YEL	EQUIPMENT GROUNDING CONDUCTOR
INSTRUMENT/SHIELDED WIRE	
BLACK	CONDUCTORS DESIGNATED AS SIGNAL NEGATIVE OR COMMON CONDUCTOR
WHITE	CONDUCTORS DESIGNATED AS SIGNAL POSITIVE

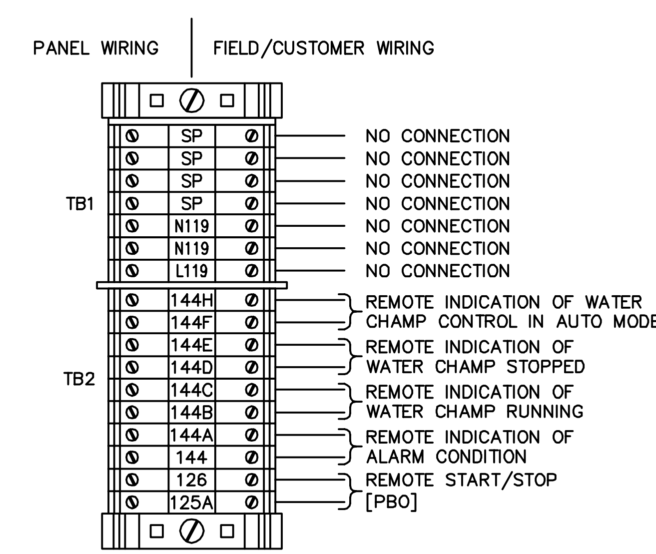


SWING-OUT PANEL



PHENOLIC MATERIAL WITH YELLOW LEGEND WITH BLACK LETTERS

SUB-PANEL



THIRD ANGLE PROJECTION

CONTROL PANEL
UL COMPLIANCE: MATERIALS AND ASSEMBLIES MANUFACTURED WITHIN SCOPE OF UNDERWRITERS LABORATORIES SHALL CONFORM TO UL STANDARDS AND HAVE AN APPLIED UL LISTING MARK.

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3-PLACE DECIMALS ±.005
FRACTIONS ±1/8"
ANGLES ±1°
DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH ANSI Y14.5M

REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN
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MANAGER		DATE	

FILE: PS-7006_3
SCALE: 0'-6"=1'-0"

TITLE: WATER CHAMP, MODEL SWCF 2 & 3HP CHEMICAL INDUCTION SYSTEM CONTROL PANEL

CLIENT

EVOQUA WATER TECHNOLOGIES
Water Technologies
Colorado Springs,
CO 719-550-2100

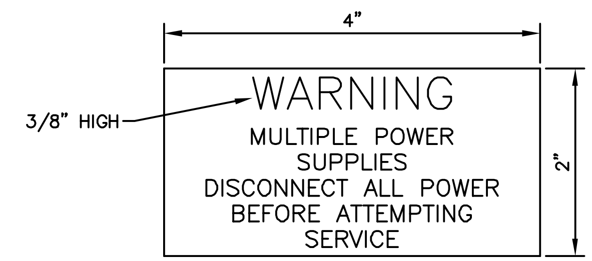
PROJECT	SAP NUMBER	DRAWING	SHEET	REV
WC	SEE TABLE 1	PS-7006	4 OF 5	3

BORDER: 22X34D PRINT NOT TO SCALE INTL REF:

BILL OF MATERIALS

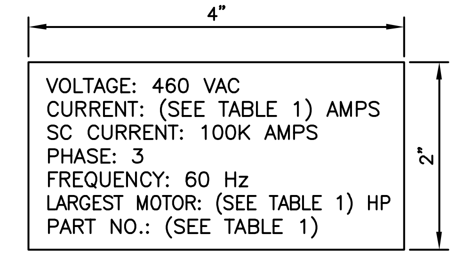
KEY NO.	DESCRIPTION	VENDOR	VENDOR PART No.	QTY	REFERENCE DESIG.
1	NAMEPLATE, EVOQUA 1" X 7.5"	EVOQUA WATER TECHNOLOGIES	W2T123953	1	
2	LABEL, WARNING	EVOQUA WATER TECHNOLOGIES	W2T15394	1	
3	CONTACTOR, MOTOR 23 AMP	ALLEN BRADLEY	100–C23D10	1	CON125
4	AUXILIARY CONTACT, MOTOR CONTACTOR, TOP MOUNT (2) N.O. & (2) N.C.	ALLEN BRADLEY	100–FA22	1	CON125
5	RELAY, 2PDT 120VAC COIL W/INDICATOR LIGHT	ALLEN BRADLEY	700–HF32A1–4	1	CR127
6	SOCKET, RELAY 2PDT BLADE TYPE	ALLEN BRADLEY	700–HN116	1	CR127
7	RETAINER CLIP, RELAY 2PDT	ALLEN BRADLEY	700–HN114	1	CR127
8	DISCONNECT SWITCH, 3 POLE 30 AMP, 600 VOLT	ALLEN BRADLEY	194R–J30–1753	1	DIS103
9	DISCONNECT SWITCH, HANDLE NEMA 4X	ALLEN BRADLEY	194R–PY	1	DIS103
10	DISCONNECT SWITCH, SHAFT	ALLEN BRADLEY	194R–S1	1	DIS103
11	METER, ELASPED TIME	ENM	T50B2	1	ETM126
12	GASKET, ELASPED TIME METER	ENM	A40047–S	1	ETM126
13	FUSE, XX AMP, 600 VOLT	MERSEN	SEE TABLE 1 ON SHT 1	3	FU104A,B,C
14	FUSE, 8/10 AMP, 600 VOLT	MERSEN	ATQR8/10	2	FU117,118
15	FUSE, 1–1/2 AMP, 600 VOLT	MERSEN	ATDR1–1/2	1	FU119
16	FUSEHOLDER, 1 POLE CLASS CC, DIN MOUNT	MERSEN	USCC1	1	FU119
17	FUSEHOLDER, 2 POLE CLASS CC, DIN MOUNT	MERSEN	USCC2	1	FU117,118
18	SWITCH, 3 POSITION SELECTOR	ALLEN BRADLEY	800H–JR2	1	HS125
19	CONTACT BLOCK, (1) N.O. & (1) N.C.	ALLEN BRADLEY	800T–XA	2	HS125
20	CONTROL, OVERLOAD, 200–480V/3PH	EVOQUA WATER TECHNOLOGIES	W2T1745	1	OLR110
21	BLOCK, MOUNTING OVERLOAD CONTROL	EVOQUA WATER TECHNOLOGIES	W2T3495	2	OLR110
22	DISTRIBUTION BLOCK, POWER, 175A, 1 IN – 4 OUT	SQUARE D	9080LBA362104	1	PD105
23	COVER, DISTRIBUTION BLOCK, POWER, 175A	SQUARE D	9080LB23	1	PD105
24	SURGE ARRESTER	SQUARE D	SDSA3650	1	SA106
25	MOUNTING BRACKET, SURGE ARRESTER	SQUARE D	QOSAMK	1	SA106
26	TRANSFORMER, 150VA	SIEMENS	MT0150A	1	T118
27	TERMINAL COVER KIT, TOUCH–SAFE SNAP–ON	SIEMENS	KTTSC4P	1	T118
28	TERMINAL BLOCK, GREY, 30 AMP 600 VOLT, TYPE UT4	PHOENIX CONTACT	3044102	17	TB1,2
29	TERMINAL BLOCK, GROUND GREEN/YELLOW, TYPE UT4–PE	PHOENIX CONTACT	3044128	2	
30	TERMINAL BLOCK END BARRIER TYPE D–UT2.5/10	PHOENIX CONTACT	3047028	1	TB1
31	TERMINAL BLOCK PARTITION PLATE TYPE ATP–UT	PHOENIX CONTACT	3047167	1	TB1,TB2
32	TERMINAL BLOCK END ANCHOR TYPE E/NS 35N	PHOENIX CONTACT	0800886	7	
33	TERMINAL BLOCK, BLANK MARKING STRIP TYPE ZB6	PHOENIX CONTACT	1051003	4	TB1,2
34	RAIL, DIN MOUNTING TYPE NS 35/7.5 PERFORATED	PHOENIX CONTACT	0801733	1	CR,TB1,2
35	LIGHT, PILOT RED LED, PUSH–TO–TEST UNIVERSAL VOLTAGE	ALLEN BRADLEY	800H–QRTH2R	1	XL128
36	LIGHT, PILOT AMBER LED, PUSH–TO–TEST UNIVERSAL VOLTAGE	ALLEN BRADLEY	800H–QRTH2A	1	XL129
37	LEGEND PLATE, AUTOMOTIVE, CUSTOM TEXT WHITE BACKGROUND	ALLEN BRADLEY	800H–W500A	3	
38	WIREWAY 1" X 2"	PANDUIT	E1X2LG6	1	
39	COVER, WIREWAY 1"	PANDUIT	C1LG6	1	
40	ENCLOSURE, 18" x 16" x 10" NEMA 4X FIBERGLASS W/WINDOW	HOFFMAN	A181610CHQRFGW	1	
41	PLATE, STEEL RELAY MOUNTING	HOFFMAN	A18P16	1	
42	SWING–OUT PANEL KIT	HOFFMAN	A18SPK16C	1	
43	GROUND LUG	THOMAS & BETTS	31003	1	
44	RUBBER GASKET TRIM	McMASTER–CARR	8507–K36	1	OLR110
45					

NOTE:
ITEM 21, OVERLOAD MOUNTING BLOCK TO BE CUT TO 1–½" HIGH



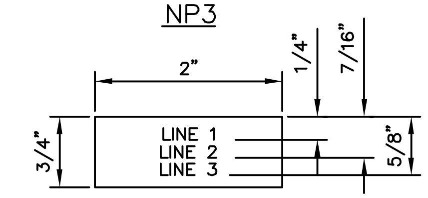
PHENOLIC MATERIAL WITH
RED LEGEND AND WHITE LETTERS
LETTER HEIGHT 3/16" HIGH

NP2



PHENOLIC MATERIAL WITH
WHITE LEGEND AND BLACK LETTERS
LETTER HEIGHT 3/16" HIGH

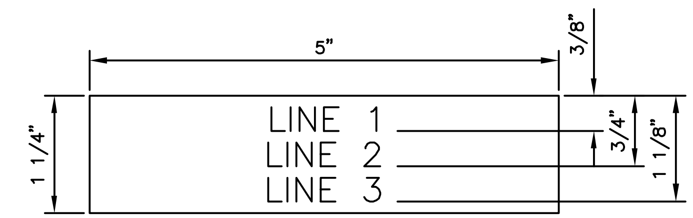
NP3



PHENOLIC MATERIAL WITH
WHITE LEGEND AND BLACK LETTERS
LETTER HEIGHT 1/8" HIGH

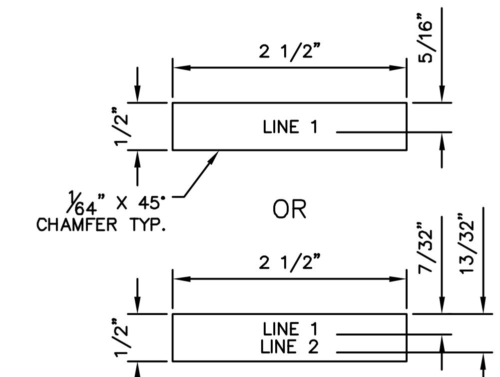
NP4–NP11

NO.		NAMEPLATE TEXT	REF
NP4	1	UL TYPE 4X	
	2	ENCLOSURE	
NP5	1	DISCONNECT	DIS103
	2	TORQUE TERMINAL	
	3	TO 12 LB–IN.	
NP6	1	GROUND LUG	
	2	TORQUE TERMINAL	
	3	TO 20 LB–IN.	
NP7	1	FU104A,B,C	FU014A
	2	REPLACE ONLY WITH	FU104B
	3	10 AMP CLASS J	FU104C
NP8	1	FU117–FU118	FU117
	2	REPLACE ONLY WITH	FU118
	3	8/10 AMP CLASS CC	
NP9	1	FU119	FU119
	2	REPLACE ONLY WITH	
	3	1–1/2 AMP CLASS CC	
NP10	1	TB1–TB2	TB1
	2	TORQUE TERMINALS	TB2
	3	TO 6.0 LB–IN.	
NP11	1	CON125	CON125
	2	TORQUE TERMINALS	
	3	TO 15.0 LB–IN.	



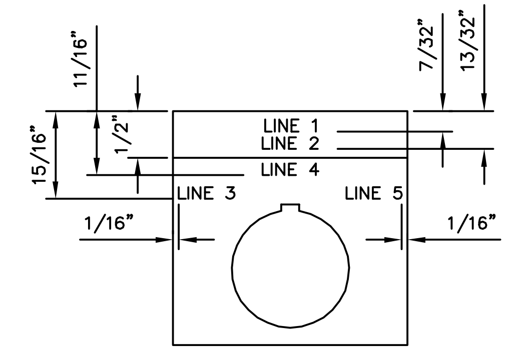
PHENOLIC MATERIAL WITH
WHITE LEGEND AND BLACK LETTERS
LETTER HEIGHT 1/4" HIGH

STYLE 1



PHENOLIC MATERIAL WITH
WHITE LEGEND AND BLACK LETTERS
LETTER HEIGHT 1/8" HIGH

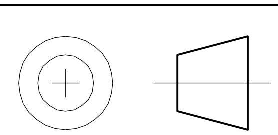
STYLE 2



ALLEN BRADLEY 800H–W500A
WHITE LEGEND WITH BLACK LETTERS
LETTER HEIGHT 1/8" HIGH

STYLE 3

NOTE: NAMEPLATE STYLES 1, 2, NP1 – NP11 ARE TO BE AFFIXED TO ENCLOSURE USING DOUBLE SIDED TAPE.



THIRD ANGLE PROJECTION

BORDER: 22X340 PRINT NOT TO SCALE

CONTROL PANEL
UL COMPLIANCE: MATERIALS AND ASSEMBLIES MANUFACTURED WITHIN SCOPE OF UNDERWRITERS LABORATORIES SHALL CONFORM TO UL STANDARDS AND HAVE AN APPLIED UL LISTING MARK.

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FILE: PS–7006_3		
SCALE: 1=1		

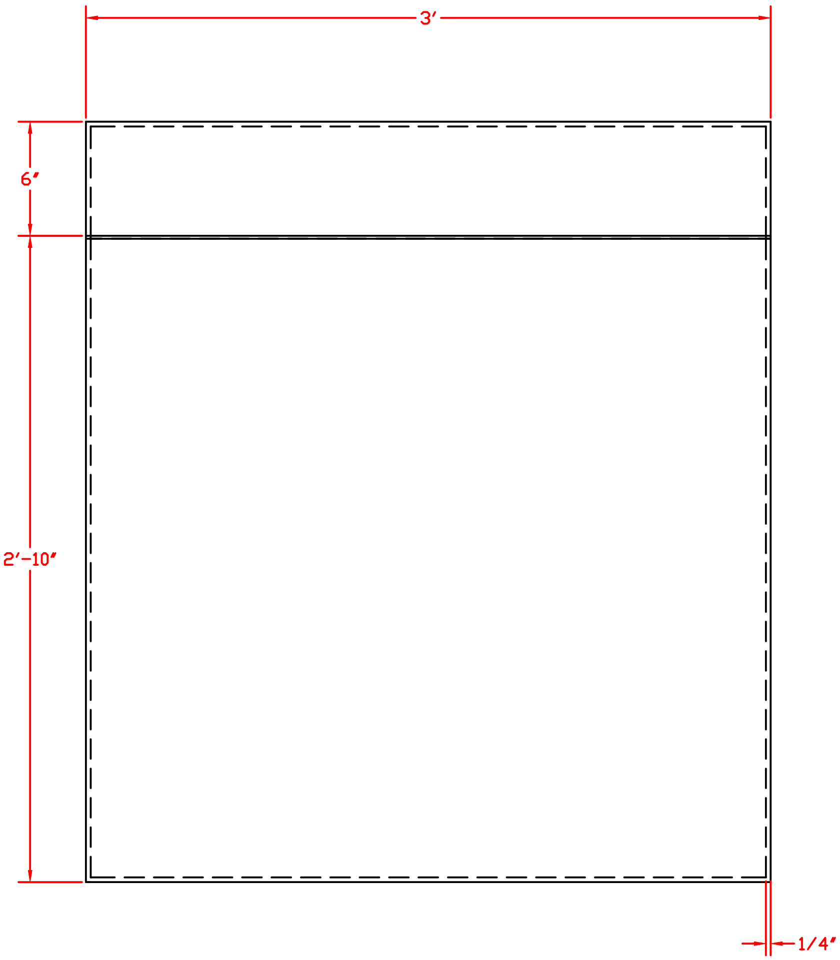
TITLE WATER CHAMP, MODEL SWCF 2 & 3HP CHEMICAL INDUCTION SYSTEM CONTROL PANEL

CLIENT

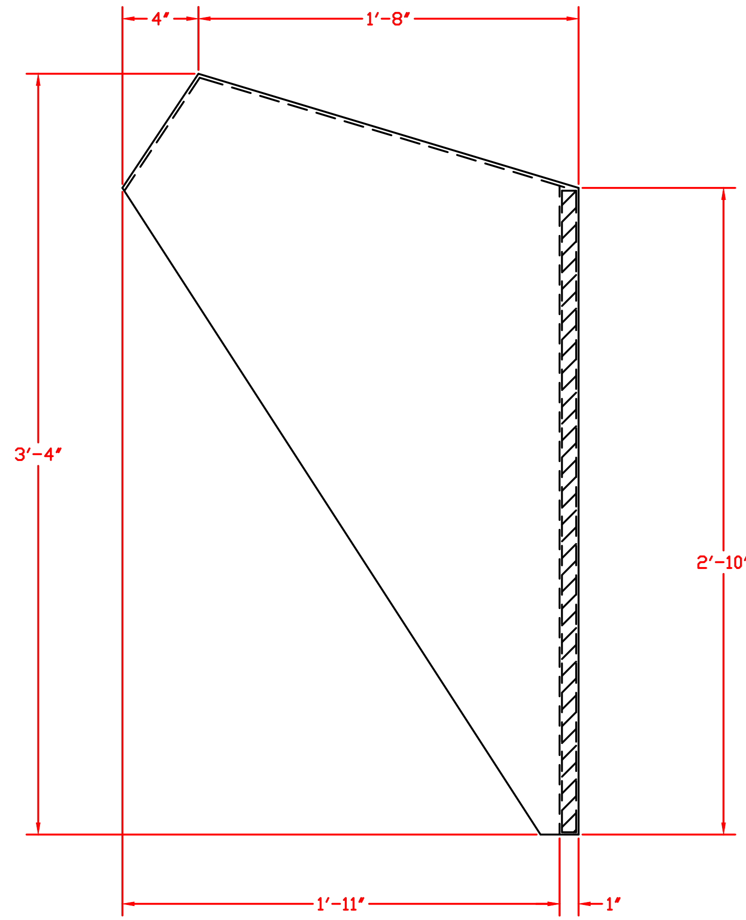
EVOQUA WATER TECHNOLOGIES
Water Technologies
Colorado Springs, CO 719–550–2100

PROJECT	SAP NUMBER	DRAWING	SHEET	REV
WC	SEE TABLE 1	PS–7006	5 OF 5	3

FEATURES: REINFORCED MOUNTING SURFACE
BLACK EDGE TRIM



FRONT ELEVATION



RIGHT ELEVATION

1) EDGE TRIM NOT SHOWN FOR CLARITY
2) ALL VALUES ARE ACCURATE TO 1/8"

TRACOM FIBERGLASS PRODUCTS	PANEL 30 INSTRUMENT PANEL SHADE, 36" W X 24" D X 40" H	
DWG. NO.	PANEL-30-01	REV 0
PROJECT	REFERENCE	DATE 2023.10.19

SHADES

TRACOM INC.

Instrument/Panel Shades

Tracom Sunshades are built to suit any situation. They shade unsheltered equipment such as Flow-Measurement devices and uncovered electronics from wind, rain, snow, debris, and sun damage. They also provide shade for viewing monitors to eliminate glare.

Instrument/Panel Shades Proctect Equipment

Panel shades keep the elements from monitoring devices. They also provide a dry environment for using any enclosed device without harm to electronics out in the field.

Customizable Shades

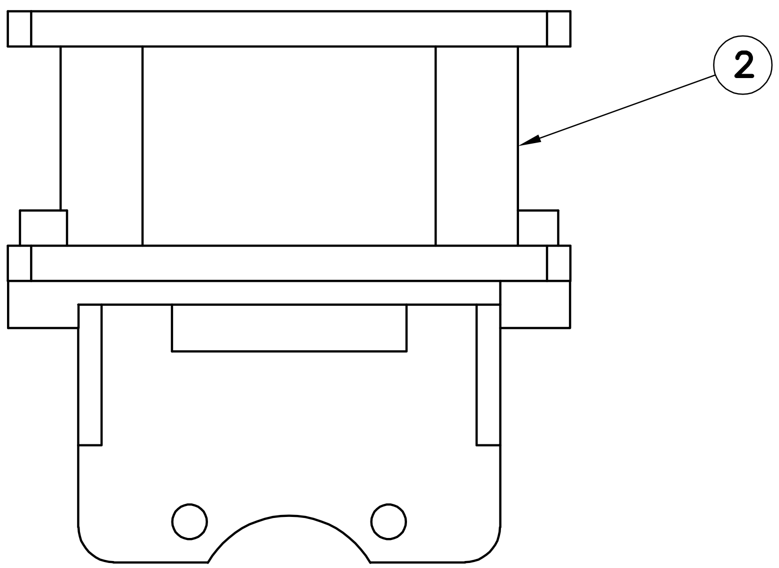
TRACOM Shades can be made in any size to suit. Standard sizes include:

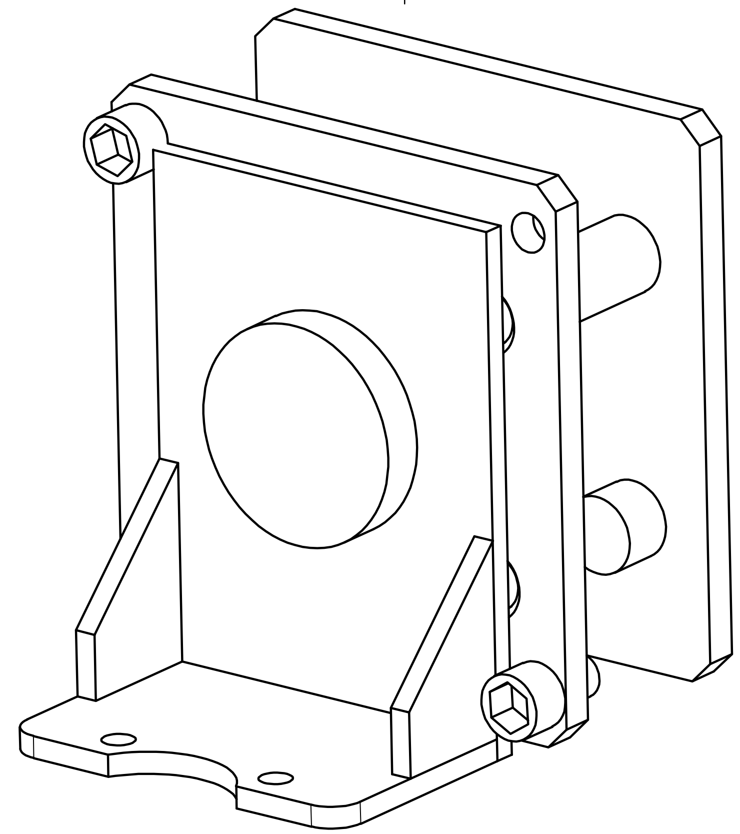
- 24" W x 15" D x 12" H
- 18" W x 12" D x 20" H
- 36" W x 24" D x 40" H

Optional Features

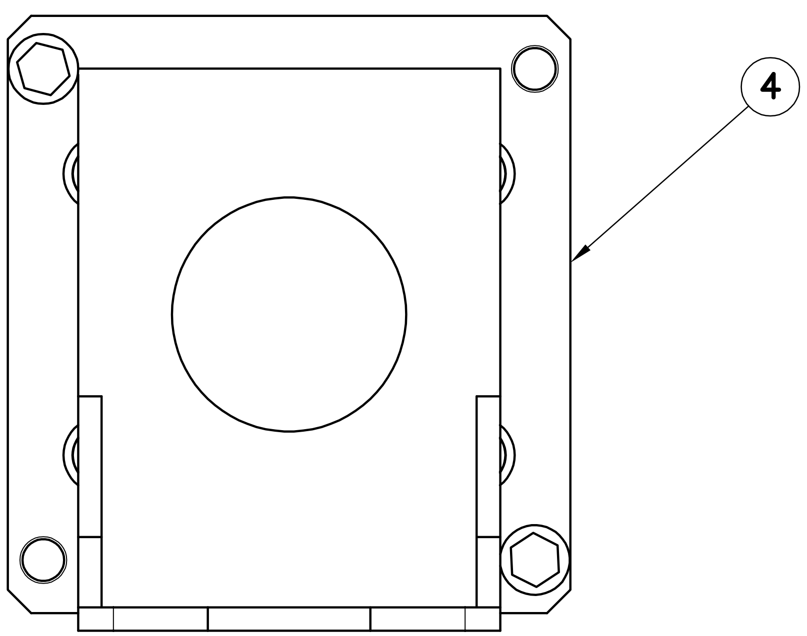
GFCI Outlet
Incandescent and Fluorescent Lighting

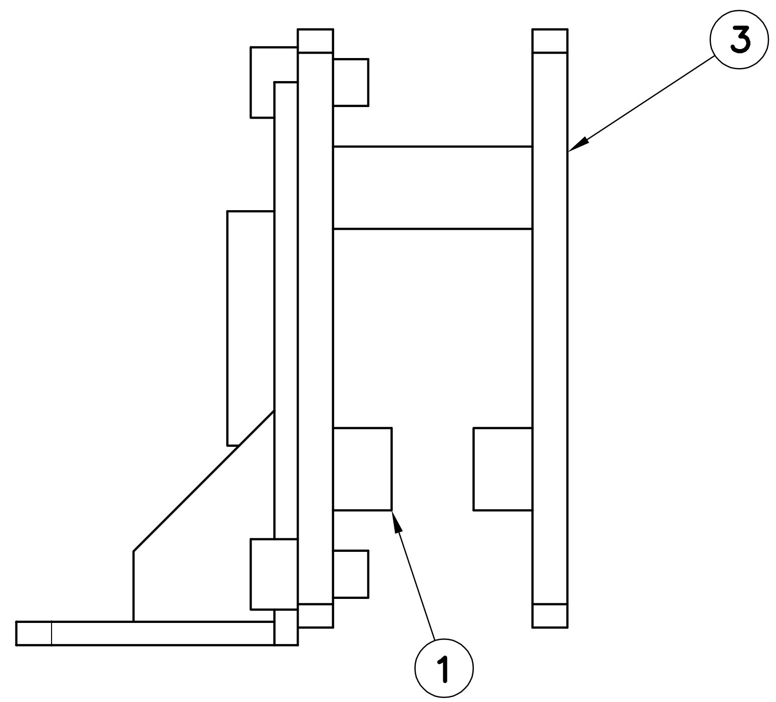
Tab F

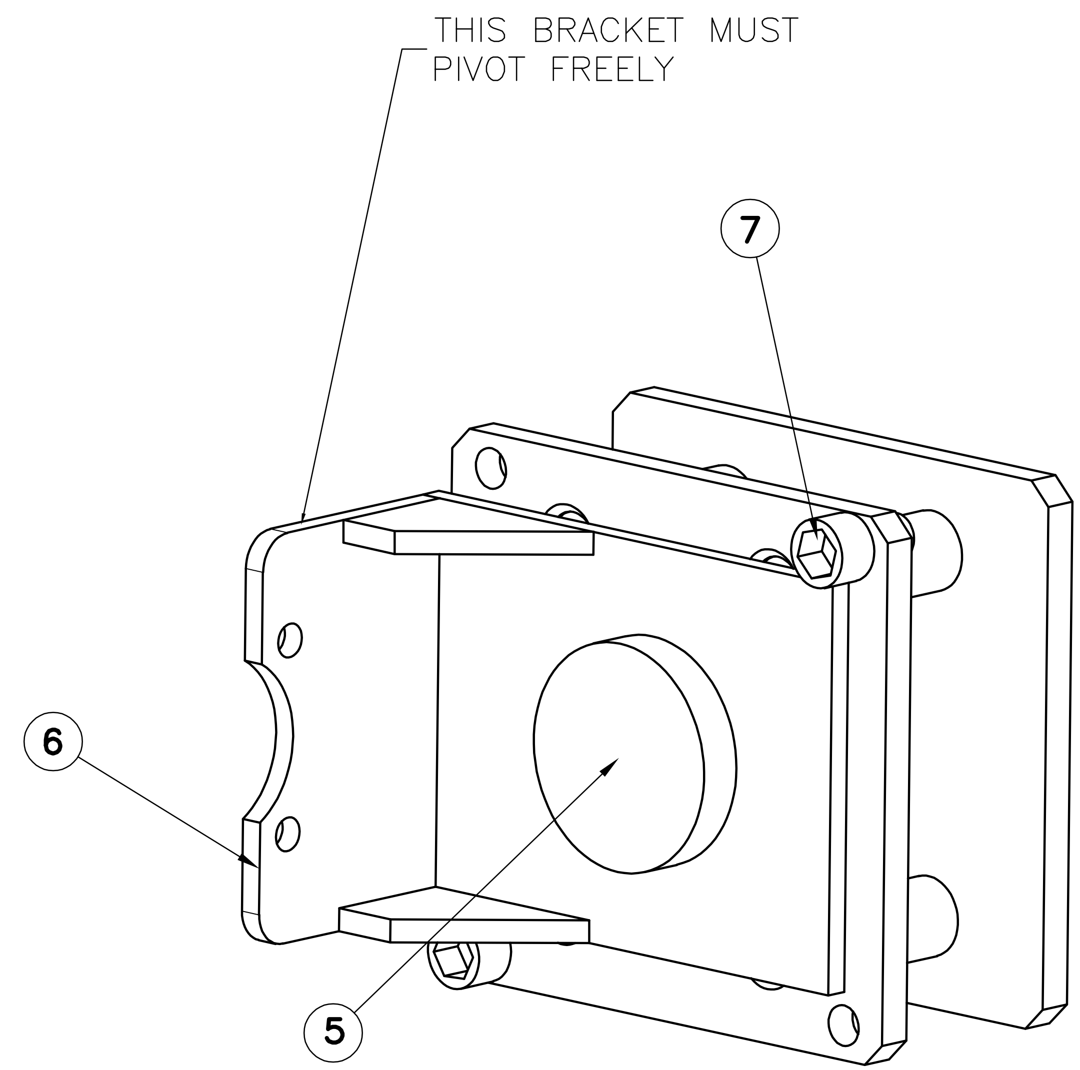




KEY	PART #	DESCRIPTION	QTY.
1	8143016	SPACER, 316 SS, .875" X 1"	4
2	8143017	SPACER, 316 SS, .875" X 2.625	2
3	6671305	PLATE, BACK SIDE OF PIVOT BRACKET, .375" THICK	1
4	6672318	PLATE, FRONT SIDE OF PIVOT BRACKET, .375" THICK	1
5	6533317	PIN FOR SWC PIVOT BRACKET	1
6	1693559	BRACKET, SWC PIVOT BRACKET, .25" THINK	1
7	7778312	SOCKET HEAD CAP SCREW, $\frac{1}{2}$–13 X $\frac{3}{4}$", 316 SS	2







LEGACY – 5950021

NOTES:

1.) MATERIAL: AS SHOWN ON THE BOM
2.) REMOVE ALL BURRS & SHARP EDGES.
3.) ALL WELDS MUST GO THROUGH PASSIVATION PROCESS
4.) TOLERANCE SHALL NOT BE CUMULATIVE
5.) QTY. 1 PER UNIT.

UNLESS OTHERWISE SPECIFIED

DIMENSIONS EXCEPT MACHINED HOLE DEPTH MARKED "DEEP"
ONE PLACE DEC. ±.050
TWO PLACE DEC. ±.010
THREE PLACE DEC. ±.005
SURFACE FINISH 125 √

MACHINED HOLE TOLERANCES	
UP TO AND INCLUDING .040" D.	+.001"/–.0005"
FROM .041"D. TO AND INCLUDING .098"D.	+.002"/–.001"
FROM .0995"D. TO AND INCLUDING .161"D.	+.003"/–.0015"
FROM .166"D. TO AND INCLUDING .250"D.	+.004"/–.002"
OVER .250"D. TO AND INCLUDING .500"D.	+.006"/–.003"
OVER .500"D. TO AND INCLUDING 1.000"D.	+.008"/–.004"
OVER 1.000"D.	+.010"/–.005"
HOLE DEPTH MARKED "DEEP"	+.025"/–.025"

MINIMUM % OF THREAD: FORMED 50%, CUT 65%
EFFECTIVE THREAD LENGTH: STRAIGHT +2/–0 PITCH, TAPERED ±1 PITCH
ANGULARITY: MACHINED ±1/2° OTHER ±1° | CONCENTRICITY: MAX. T.I.R. .005"
BREAK ALL SHARP EDGES (MAX RAD OR CHAMFER): EXTERNAL .015" INTERNAL .035"

MATERIAL: 316 SS

TITLE: ASSEMBLY DRAWING
UNIVERSAL SWIVEL BRACKET
4" SWC WATER CHAMP

REV.	CHANGE No.	DATE
2	31094	01DEC11

SCALE: 1:1 | DR'N: . | CH'K: .
RELEASED: . | ENG: . | Q.A.: .

xylem

NUMBER: W3T108026

THIRD ANGLE PROJECTION

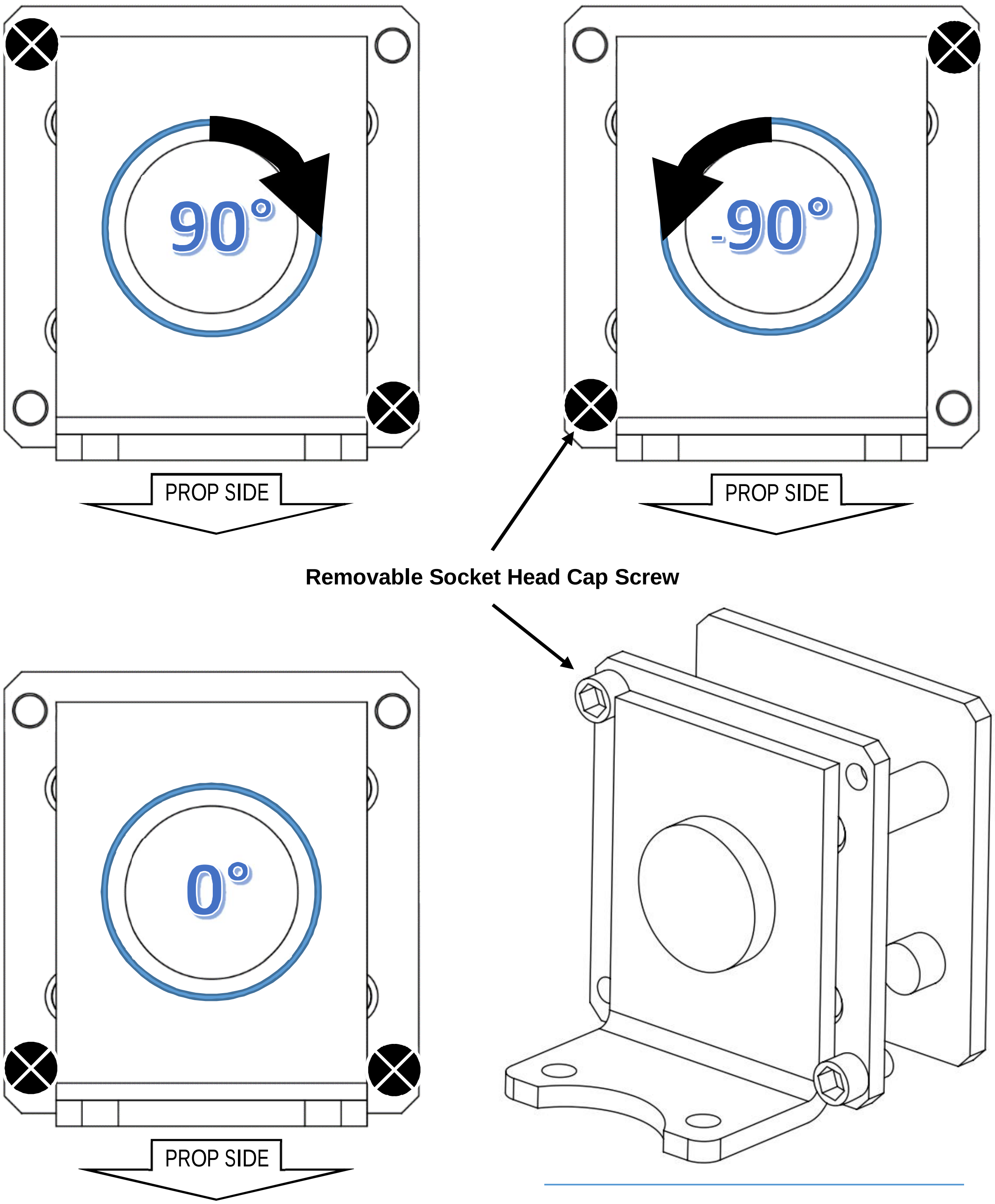
DESIGN CENTER: USA | SHEET 1 OF 3

MODEL: . DWG: W3T108026–0102

TOLERANCES TO F937.400 | D |

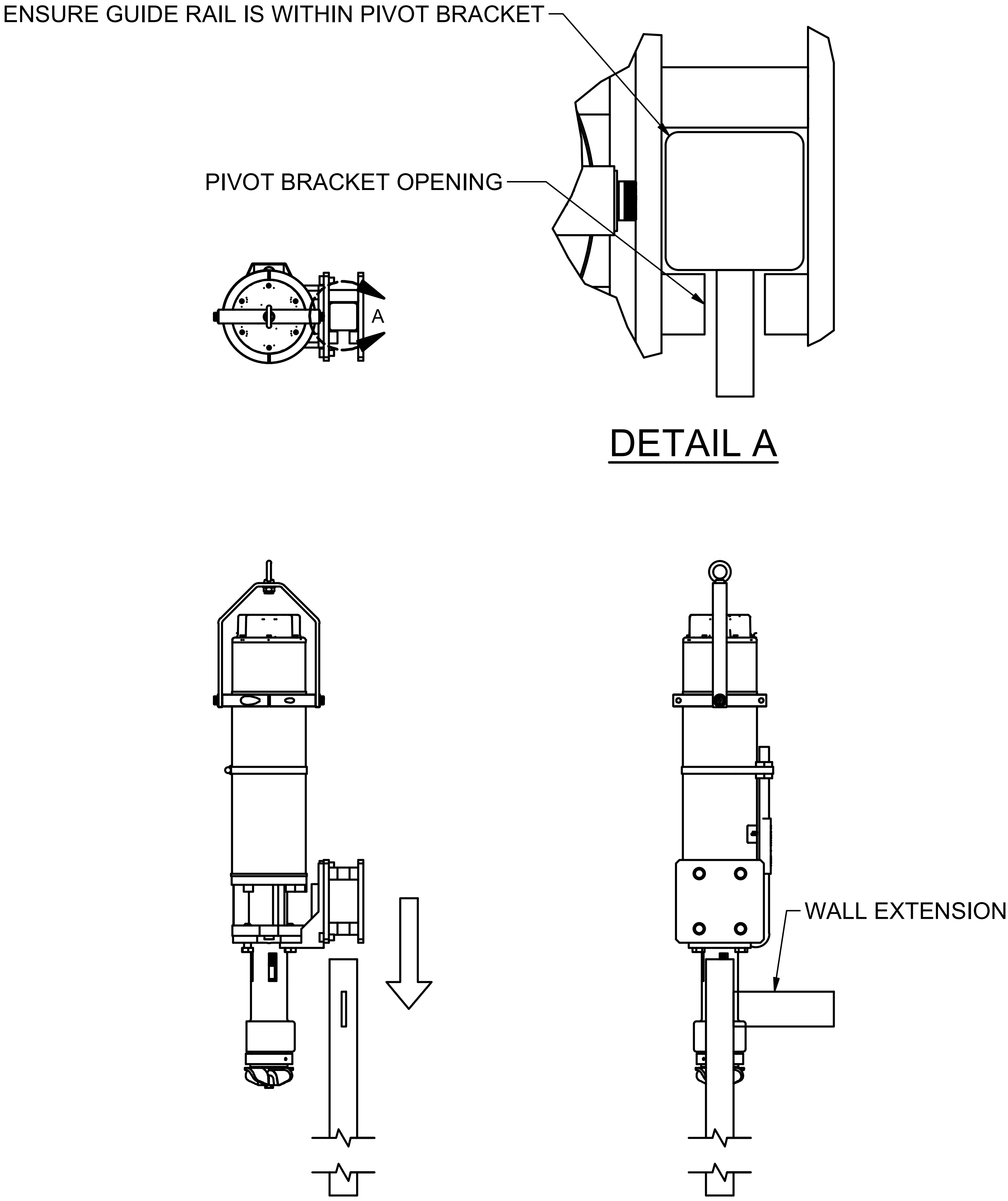
Wallace & Tiernan®
an evoqua brand

Water Champ® Pivot Bracket Configurations



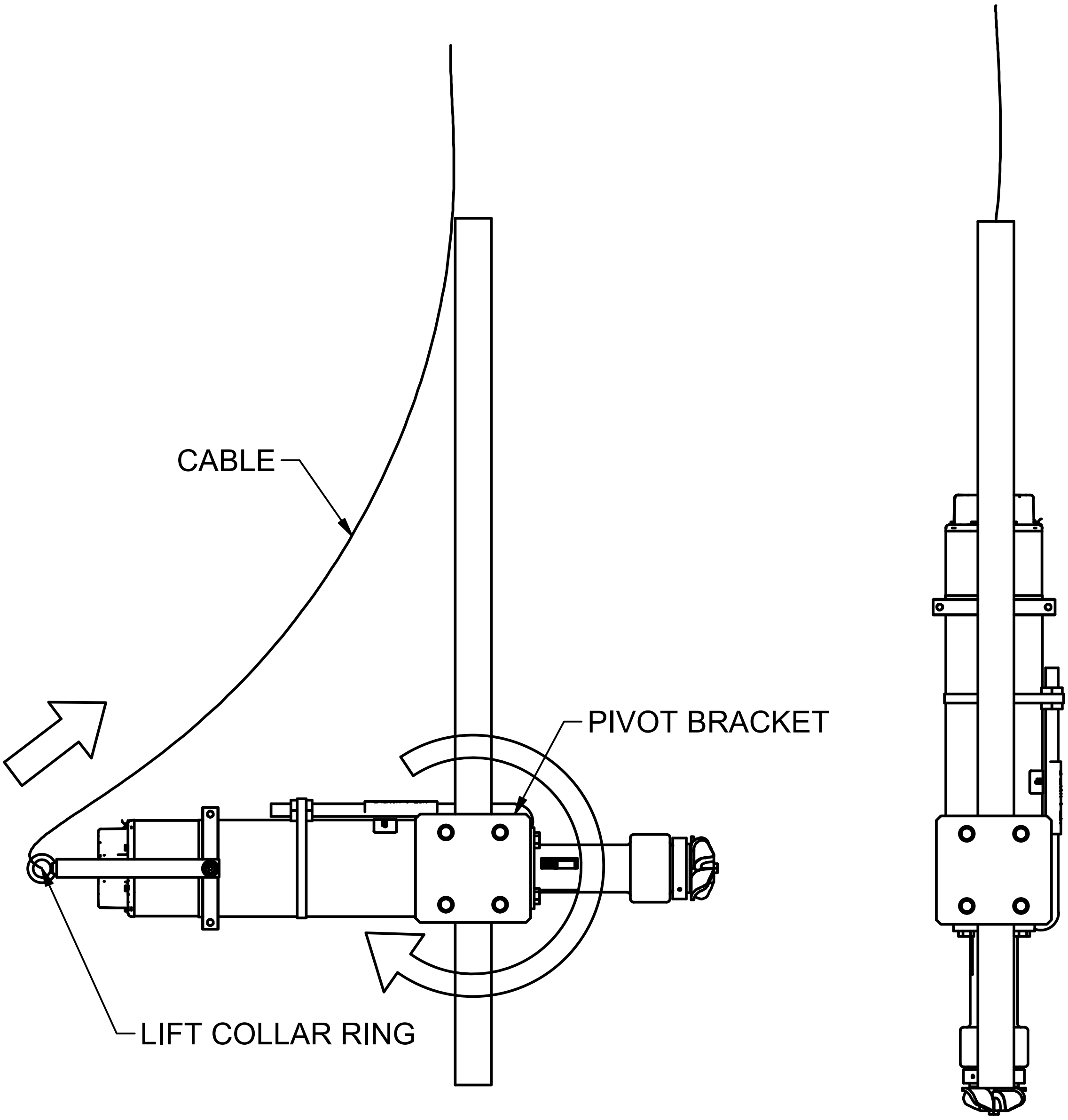
Isometric display is of 90° position

***FX Water Champ Shown**



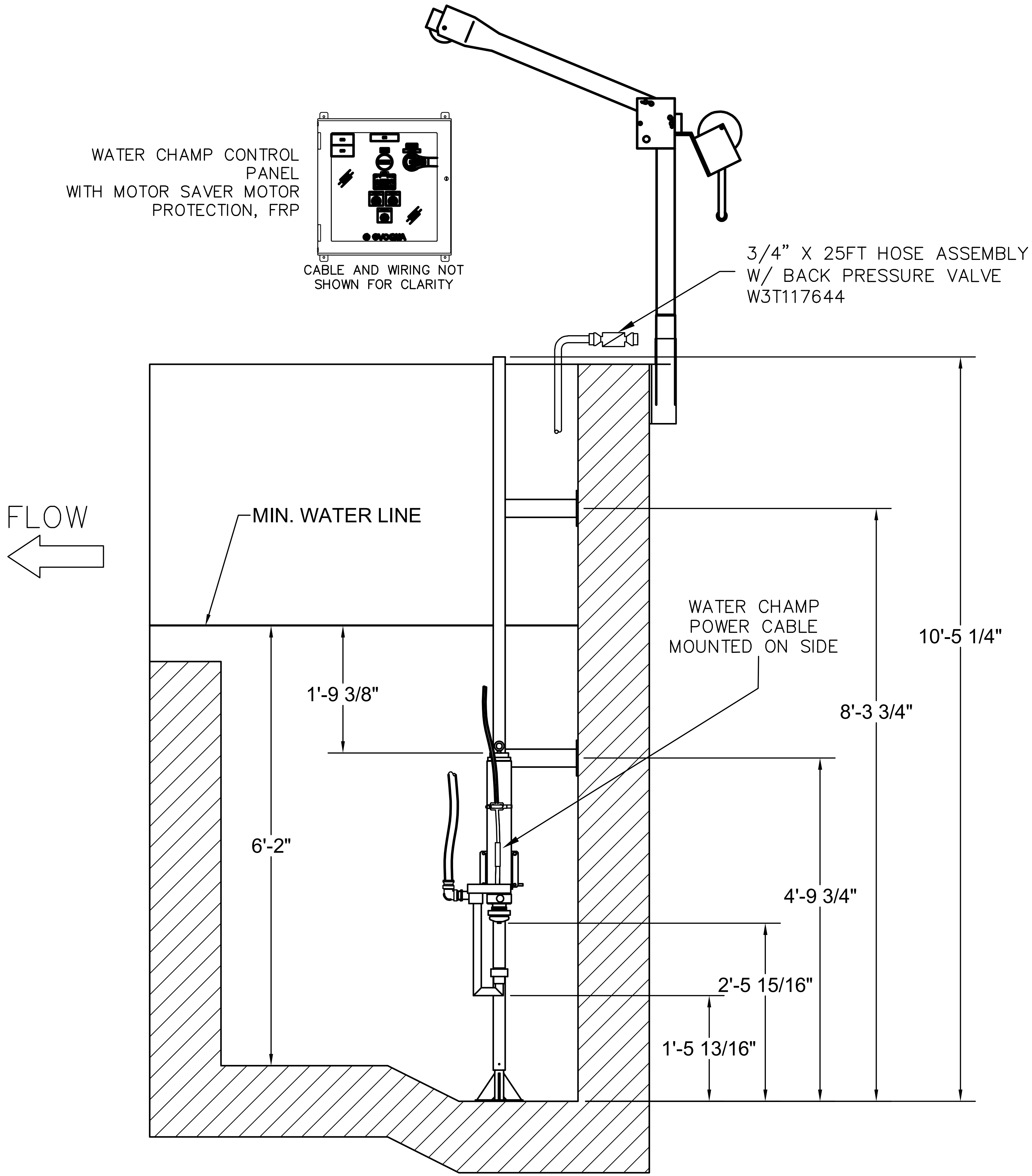
1. ENSURE PIVOT BRACKET OPENING IS TOWARDS THE WALL EXTENSION
2. LOWER WATER CHAMP AND MANEUVER THE RAIL THROUGH THE PIVOT BRACKET
3. CONTINUE LOWERING WATER CHAMP UNTIL IT IS FULLY SETTLED

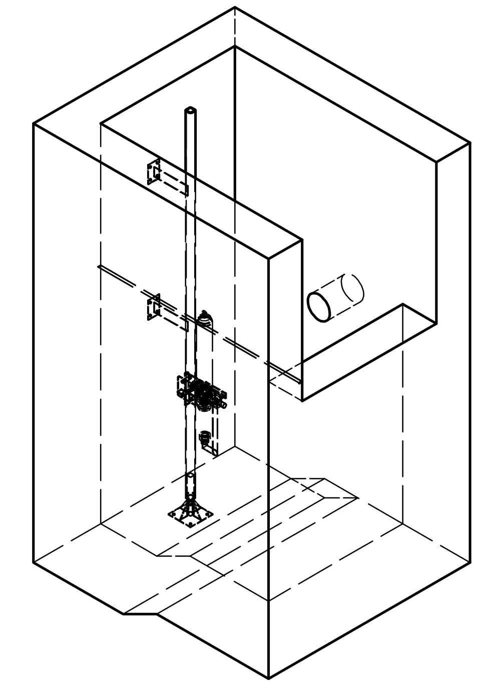
FX Water Champ Shown



1. CRANE/WINCH LIFTS UP CABLE AND PULLS THE RING ON PULL LIFT COLLAR
2. LIFT COLLAR PULLS ON REAR OF WATER CHAMP
3. WATER CHAMP ROTATES AROUND THE PIVOT BRACKET INTO VERTICAL POSITION

Tab G





NOTES:

1. UNIT MUST BE COMPLETELY SUBMERSED AT LEAST 18 INCHES DURING OPERATION.

2. CABLES & HOSES NOT SHOWN IN ALL VIEWS FOR CLARITY.

3. CONTROL PANEL LOCATION TO BE DETERMINED BY OTHERS

DWN BY: D.ACUNA	DATE 9/4/2022	TITLE GENERAL ARRANGEMENT ASSEMBLY
CKD BY:	DATE	
APPD BY:	DATE	CLIENT NAN INC.
MGD BY:	SCALE NTS	xylem XYLEM

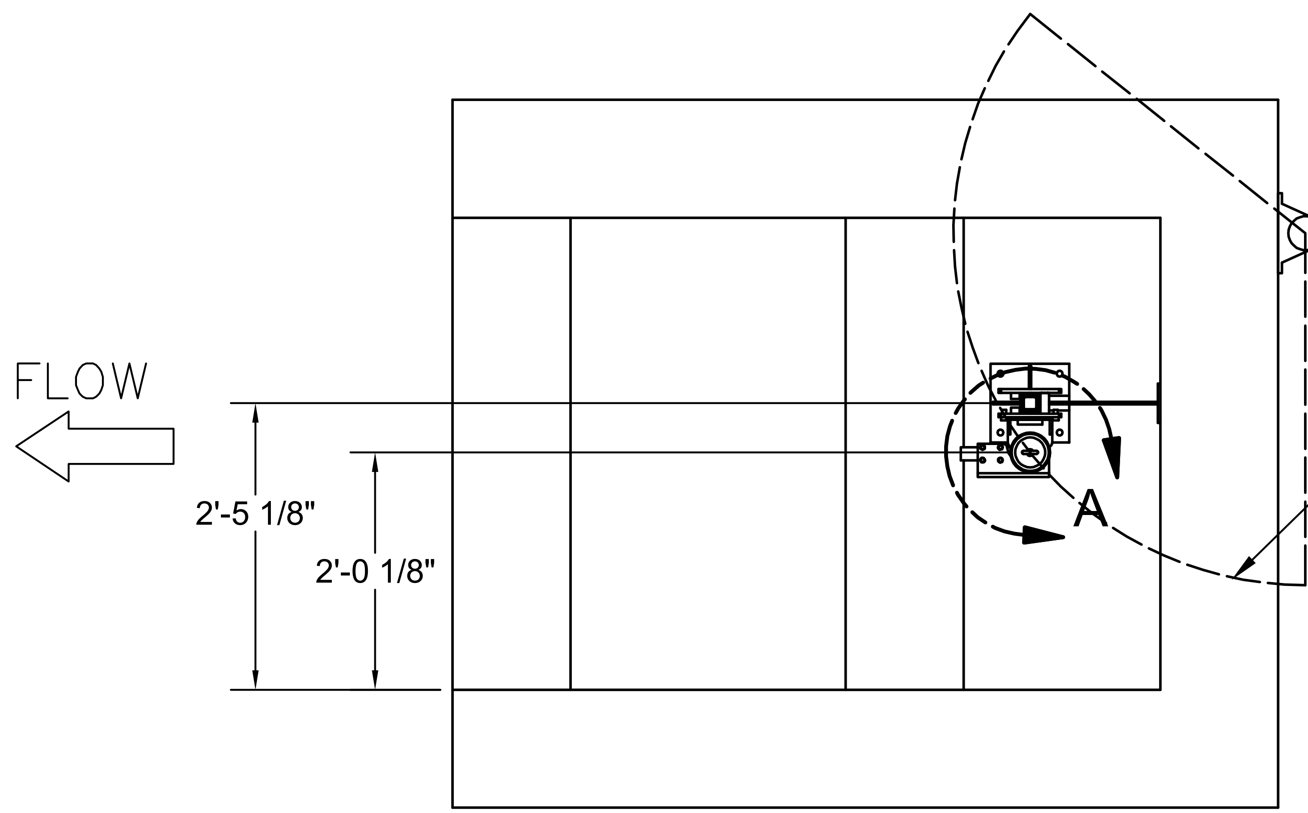
PART NUMBER	PROJ/PROD NUMBER	CODE	FILE/DRAWING NUMBER	SHEET	REV
TBD	1619831		Standard	1 OF 3	1

STD:11x17_B v5.00 BAR = 1" AT PLOT SCALE

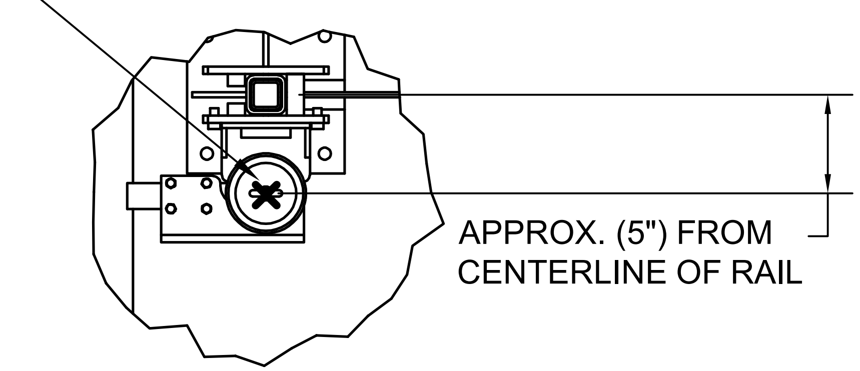
NOTES: (CONT.)

4. (IMPORTANT) LOCATION OF DAVIT CRANE BASE TO BE DETERMINED BY OTHERS. SWING ARM REQUIREMENT TO MATCH CENTERLINE OF WATER CHAMP

5. CRANE MUST BE ABLE TO LOWER AND LIFT WATER CHAMP WITHOUT BINDING ON RAIL. CRANE CABLE SHOULD DROP APPROX. 5" FROM CENTERLINE OF RAIL. SEE "DETAIL A".



TOP VIEW



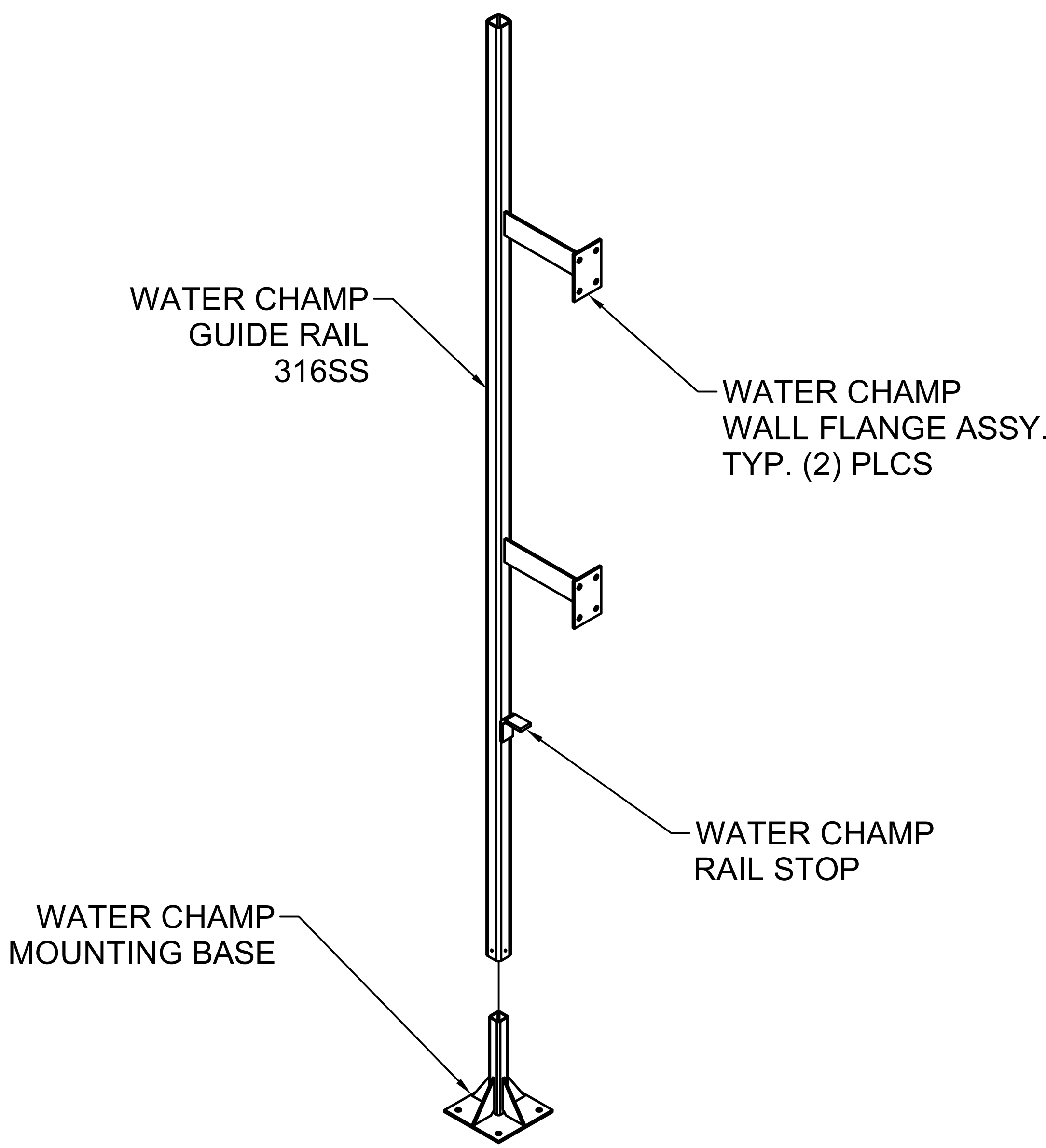
DETAIL A

DWN BY:	DATE	TITLE
D.ACUNA	9/4/2022	GENERAL ARRANGEMENT ASSEMBLY
CKD BY:	DATE	CLIENT NAN INC.
APPD BY:	DATE	XYLEM
MGD BY:	SCALE NTS	

PART NUMBER	PROJ/PROD NUMBER	CODE	FILE/DRAWING NUMBER	SHEET	REV
TBD	1619831		Standard	2 OF 3	1

STD:11x17_B v5.00

BAR = 1" AT PLOT SCALE



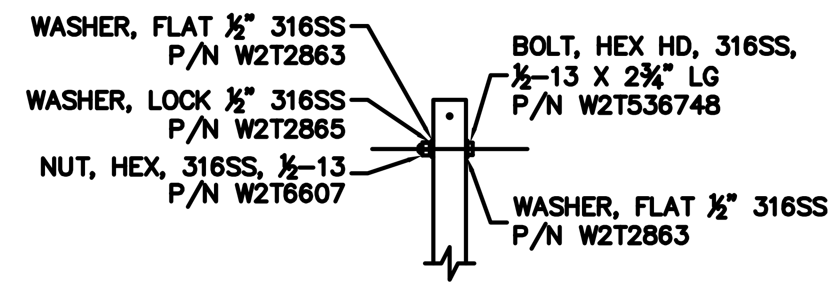
NOTES (CONT):

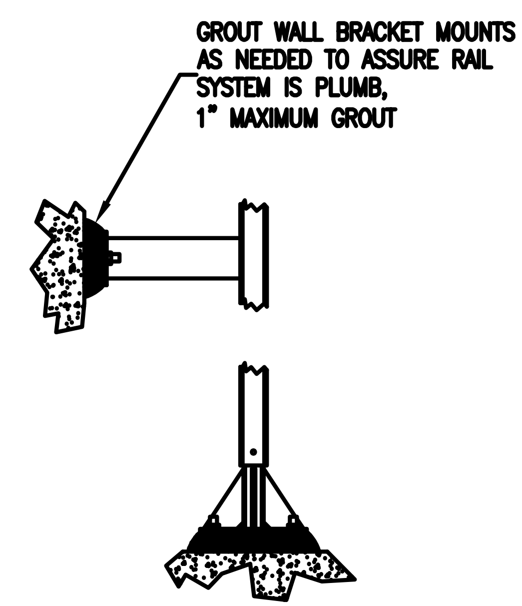
6. CONTRACTOR RESPONSIBLE FOR THE FOLLOWING:

A. ASSURING THE RAIL SYSTEM PLUMB AND THAT THE MOTOR TRAVERSES THE RAIL SYSTEM FREE FROM BINDING.

B. CUTTING, MODIFYING OR RELOCATING HANDRAILS AS NEEDED TO ALLOW ACCESS TO THE WATER CHAMP HOIST.

C. INFORMATION REGARDING ELEVATIONS, MOUNTING SURFACES AND OBSTRUCTIONS MUST BE VERIFIED TO INSURE PROPER INSTALLATION.





DWN BY: D.ACUNA	DATE 9/4/2022	TITLE GENERAL ARRANGEMENT ASSEMBLY
CKD BY:	DATE	
APPD BY:	DATE	CLIENT NAN INC.
MGD BY:	SCALE NTS	xylem XYLEM

PART NUMBER	PROJ/PROD NUMBER	CODE	FILE/DRAWING NUMBER	SHEET	REV
TBD	1619831		Standard	3 OF 3	1

BAR = 1" AT PLOT SCALE

STD:11x17_B v5.00

Tab H

SEISMIC CALCS TO FOLLOW